

An Explainer on Commutative Algebra

A learning path from the modern, categorical point of view

Will Lancer

`will.m.lancer@gmail.com`

Commutative algebra is two subjects wearing one coat. It is the *local algebra of geometry*—the thing that tells you what a variety looks like near a point—and it is the *arithmetic of numbers*—the thing that tells you how primes factor in a ring of integers. The modern, Grothendieck-era insight is that these are the *same* subject, because a commutative ring *is* a space: its points are prime ideals, its functions are its elements, and a map of rings is a map of spaces running the other way. These notes are a guided path to that point of view. We learn the subject the way it is actually organized today: **Spec** (the spectrum) as the central object, **modules** and **localization** as the two everyday tools, and the language of **categories, universal properties, and adjoint functors** as the connective tissue that makes the whole thing hang together.

What this document is. It is the **recommended learning path**—an annotated table of contents. Each section says *what* it covers, *why* it sits where it does in the modern arc, and *what to read* for it in the three primary sources (see the reference list). Read it as a syllabus and a map; the full worked notes are built out section by section on top of this skeleton.

Prerequisites. A first course in abstract algebra (groups, rings, fields), linear algebra, and a little point-set topology. *No* category theory is assumed—we build it from scratch in §2, exactly where it is first needed.

How to read. Boxes labelled Definition/Theorem are the load-bearing statements; Intuition and Note boxes are the “how a working algebraist pictures it” commentary; yellow Equation boxes hold a “carry this in your head” slogan. Green Exercise boxes are scattered throughout, tagged (**easy**), (**classic**), (**medium**), or (**thinker**)—do them as you go. An end-of-path exercise set and an annotated “books to own” list close the notes.

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1 The big picture: commutative algebra as geometry and arithmetic

Before a single definition, we answer the only question that matters at the start: why does this subject exist, and what is it secretly about? The answer is a dictionary. To every commutative ring R we attach a space $\text{Spec } R$ whose points are the prime ideals of R ; the elements of R become functions on that space, and a homomorphism $R \rightarrow S$ becomes a continuous map $\text{Spec } S \rightarrow \text{Spec } R$. Geometry (varieties, their singularities, families) and arithmetic (factorization, ramification, class groups) turn out to be two readings of this one dictionary. This opening section sets up the dictionary, names the three tools we will lean on the whole way through, and lays out the route.

1.1 Two faces of one subject: the local algebra of geometry, the arithmetic of numbers

Start with two questions you can already feel the weight of, one from geometry and one from arithmetic.

The geometric question. Take a plane curve, say $C: y^2 = x^3 - x$. How do you study it *algebraically*? The natural move is to look at the polynomial functions on the plane and see which two agree on C : two polynomials restrict to the same function on C exactly when their difference is a multiple of $y^2 - x^3 + x$. So the functions on C form the quotient ring $k[x, y]/(y^2 - x^3 + x)$. Every geometric feature of C —where it is smooth, where it crosses itself, how many pieces it has—turns out to be readable off this ring. Geometry has become algebra.

The arithmetic question. In $\mathbb{Z}$ every integer factors uniquely into primes. Now enlarge $\mathbb{Z}$ to $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$. There,

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

and all four factors are irreducible: unique factorization *fails*. Dedekind’s resolution—one of the births of the subject—was to stop factoring *elements* and start factoring *ideals*, in which language (6) does factor uniquely, and to measure the failure by an invariant of the ring (the class group). Arithmetic has become algebra about ideals.

The point is that both questions are about a **commutative ring**: in the first it is a ring of *functions on a space*, in the second a ring of *numbers*. Commutative algebra is the claim that these are not two subjects but one.

Idea 1.1 (The thesis of commutative algebra)

A commutative ring is, simultaneously, the “local algebra of geometry” (the functions on a space) and the “arithmetic of a number system.” The two readings are unified because both are commutative rings, and the *same* tools—ideals, modules, localization—resolve both the geometric and the arithmetic questions. Learning to read a ring in both dialects at once is the whole skill.

The unification is not vague analogy; it is a precise dictionary, and the cleanest place to see it is the parallel between the integers $\mathbb{Z}$ and the polynomial ring $k[x]$ in one variable over a field k .¹ Both are **principal ideal domains**: every ideal is generated by one element. In $\mathbb{Z}$ the “irreducible” objects are the prime numbers p ; in $k[x]$ (take k algebraically closed for a clean statement) they

¹This $\mathbb{Z} \leftrightarrow k[x]$ parallel is the backbone of the whole modern viewpoint; A–M open their book by announcing that the “central notion in commutative algebra is that of a prime ideal” and that, following the modern trend, they emphasize modules and localization (A–M, Introduction). We take that emphasis as our organizing principle.

are the linear polynomials $x - a$, one for each point a of the line. Read the table across, and the arithmetic column and the geometry column are the same story told twice.

	Arithmetic: $\mathbb{Z}$	Geometry: $k[x]$
the ring	integers $\mathbb{Z}$	polynomials $k[x]$
an irreducible	prime number p	linear factor $x - a$
a “point”	prime ideal (p)	prime ideal $(x - a)$
residue field there	$\mathbb{Z}/(p) = \mathbb{F}_p$	$k[x]/(x - a) = k$
“value” of an element	$n \bmod p$	$f(a)$
the generic point	(0) , with field $\mathbb{Q}$	(0) , with field $k(x)$

The last two rows are the punchline, and they are worth pausing on. Reducing an integer n modulo a prime p and evaluating a polynomial f at a point a are the same operation: in each case you take an element of the ring and pass to the quotient by the “point” to read off its value there. An integer is, in this precise sense, a function on the prime numbers, whose value at p is $n \bmod p$. That single realization—numbers are functions, primes are points—is the seed from which the entire modern subject grows. The next subsection makes it a definition.

A good analogy earns its keep by showing you exactly where it frays, so before we lean on this one we should say where it does.

Remark 1.1 (Where the analogy frays—and the deeper analogy underneath). The $\mathbb{Z} \leftrightarrow k[x]$ dictionary is a guide, not an identity, and its seams are themselves famous. Two honest disanalogies. *First*, the residue fields of $k[x]$ are all the same field k , whereas those of $\mathbb{Z}$ are the *varying* finite fields $\mathbb{F}_p$ —so $k[x]$ is really the *geometric* (“equal characteristic”) analogue of $\mathbb{Z}$. The closer *arithmetic* analogue is the polynomial ring $\mathbb{F}_q[t]$ over a finite field, whose residue fields $\mathbb{F}_q[t]/(\pi)$ are again finite. *Second*, $\mathbb{Z}$ secretly carries one extra “point”: the ordinary absolute value on $\mathbb{Q}$ is a further, **archimedean** place with no algebraic counterpart, so $\text{Spec } \mathbb{Z}$ behaves like an *affine* curve that is missing its point at infinity—supplying it is the starting gun of Arakelov geometry. Far from being a defect, the full analogy

$$\{\text{number fields}\} \longleftrightarrow \{\text{function fields of curves over a finite field } \mathbb{F}_q\},$$

with a ring of integers $\mathcal{O}_K$ playing the role of (the affine part of) a curve, is the backbone of modern number theory: it is the setting of Weil’s proof of the Riemann hypothesis for curves and of the function-field Langlands program of Drinfeld and L. Lafforgue. We will keep meeting it.

1.2 The central dogma: rings are spaces, primes are points, functions are elements

Here is the dictionary in full. The precise definitions and proofs come in §3; right now we want the shape of the idea so that everything afterward has a place to land.

Preview 1.1 (The ring–space dictionary (to be made precise in §3))

To a commutative ring R we attach a space, the **spectrum** $\text{Spec } R$, whose *points* are the **prime ideals** of R .

- An element $f \in R$ is a *function* on $\text{Spec } R$. Its *value* at a point $\mathfrak{p}$ is the image of f in the

residue field $\kappa(\mathfrak{p}) \stackrel{\text{def}}{=} \text{Frac}(R/\mathfrak{p})$ —the fraction field of the (integral domain) quotient $R/\mathfrak{p}$.^a In particular, f **vanishes at** $\mathfrak{p}$ (its value there is 0) exactly when $f \in \mathfrak{p}$, so the points where f is zero are precisely the primes that contain it—this is what will turn “solving $f = 0$ ” into the theory of ideals.

- A ring homomorphism $\varphi: R \rightarrow S$ induces a map of spaces in the *opposite* direction, $\text{Spec } S \rightarrow \text{Spec } R$, sending a prime $\mathfrak{q} \subseteq S$ to its preimage $\varphi^{-1}(\mathfrak{q}) \subseteq R$.

^a $R/\mathfrak{p}$ is a domain precisely because $\mathfrak{p}$ is prime; that is the whole reason primes, not arbitrary ideals, are the points. See §3.

Two design decisions in that box deserve their motivation, because each is a place where a beginner’s first guess is wrong.

Why prime ideals, and not just maximal ones? The classical points of a space—a single number a , a single point of the plane—correspond to *maximal* ideals (this is the Nullstellensatz, §10). So why widen “point” to all primes? Two reasons, one formal and one geometric. Formally: the preimage of a prime ideal under a ring map is always prime, but the preimage of a *maximal* ideal need not be maximal.² If we want Spec to be a *functor*—so that algebra translates to geometry automatically—we are forced to use primes. There is also a vivid reason the *prime* condition is the right one. The defining property $fg \in \mathfrak{p} \Rightarrow f \in \mathfrak{p} \text{ or } g \in \mathfrak{p}$ says, in the language of values, that a product of functions can vanish at $\mathfrak{p}$ only if one of the factors already does; the vanishing at $\mathfrak{p}$ cannot be split into two independent pieces, so $\mathfrak{p}$ behaves like an *irreducible* locus—and, dually, its ring of functions $R/\mathfrak{p}$ has no zero-divisors. “Prime ideal” is the algebra of “irreducible piece of the space.” Geometrically: a non-maximal prime is the **generic point** of an irreducible subspace. In $k[x, y]$ the prime $(y - x^2)$ is not a single point of the plane; it is the entire parabola $y = x^2$, packaged as one point of $\text{Spec } k[x, y]$. Allowing irreducible subvarieties to *be* points is one of the most useful moves in all of geometry, and it is free once you take primes seriously.

Why does the map of spaces go backwards? Because functions pull back. If $\varphi: R \rightarrow S$ presents $\text{Spec } S$ as a space mapping to $\text{Spec } R$, then a function on the target ($f \in R$) becomes a function on the source ($\varphi(f) \in S$) by composition. Pulling functions back is contravariant, so the spaces map the opposite way to the rings. This is the same variance you already know from topology: a continuous map $X \rightarrow Y$ induces $C(Y) \rightarrow C(X)$ on rings of continuous functions.

Equation 1.1 (The slogan to carry in your head)

$$\underbrace{\text{value of } f \text{ at the point } \mathfrak{p}}_{\text{image of } f \text{ in } \kappa(\mathfrak{p})} \rightsquigarrow \underbrace{n \bmod p}_{\text{in } \mathbb{Z}} = \underbrace{f(a)}_{\text{in } k[x]}.$$

Elements are functions; primes are points; reduction modulo a prime is evaluation.

Example 1.1 (The two spectra to know by heart)

(a) **$\text{Spec } \mathbb{Z}$.** The primes of $\mathbb{Z}$ are (0) and (p) for each prime number p . The points (p) are “closed” (ordinary points), with residue field $\kappa((p)) = \mathbb{F}_p$; the value of n there is $n \bmod p$. The

²Example: $\mathbb{Z} \hookrightarrow \mathbb{Q}$. The ideal (0) is maximal in the field $\mathbb{Q}$, but its preimage $(0) \subseteq \mathbb{Z}$ is prime and not maximal. So “maximal ideals” do not form a functorial notion of point, while primes do. This is exactly why Spec uses primes.

point (0) is the *generic point*: its residue field is $\kappa((0)) = \mathbb{Q}$, and the value of n there is n itself. So $\text{Spec } \mathbb{Z}$ is the prime numbers together with one extra “smeared-out” point sitting over all of them.

(b) **$\text{Spec } k[x]$, $k = \bar{k}$.** The primes are (0) and $(x - a)$ for $a \in k$. The closed points $(x - a)$ are the line k ; the residue field is k and the value of f at $(x - a)$ is $f(a)$ —honest evaluation. Again there is one generic point (0) , with residue field $k(x)$, the field of rational functions.

Intuition 1.1 (Elements are functions, but stranger)

An element behaves like a function on $\text{Spec } R$, with two twists that ordinary functions lack. First, its values live in *different fields at different points* (the residue field $\kappa(\mathfrak{p})$ changes with $\mathfrak{p}$), so “ f ” is a function whose codomain varies. Second, an element can be nonzero yet vanish at points, and—most strikingly—a nonzero **nilpotent** can vanish at *every* point: in $k[x]/(x^2)$ the element x is not 0, but its value in every residue field is 0. Classical functions cannot do this; the ring remembers an “infinitesimal” direction that the naive set of points forgets. We keep nilpotents (rather than quotient them away) precisely because they record this infinitesimal data—the ring $k[\varepsilon]/(\varepsilon^2)$ is a point carrying a tangent vector. This is the first hint of why *scheme* theory works with rings, not just point sets.

The contravariant map of spectra is not a technicality; it is where arithmetic turns into the geometry of covering maps.

Example 1.2 (A ring map as a branched cover: $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$)

The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$ of the integers into the Gaussian integers induces $\text{Spec } \mathbb{Z}[i] \rightarrow \text{Spec } \mathbb{Z}$. Over a point (p) of $\text{Spec } \mathbb{Z}$ sits the *fiber*: the primes of $\mathbb{Z}[i]$ lying above p , i.e. the way p factors in $\mathbb{Z}[i]$. And the classical trichotomy of number theory is exactly a statement about this map:

$$(5) = (2 + i)(2 - i) \text{ (splits: two points), } (3) \text{ (inert: one point), } (2) = -i(1 + i)^2 \text{ (ramifies).}$$

So $\text{Spec } \mathbb{Z}[i] \rightarrow \text{Spec } \mathbb{Z}$ is a (branched) double cover^a of $\text{Spec } \mathbb{Z}$, and “how p splits” is the same data as “what the fiber looks like.” The arithmetic of the Gaussian integers *is* the geometry of this covering map—the prototype of everything in algebraic number theory.

^aThe cover is branched over only finitely many primes: 2 is the sole *ramified* prime here, and it is exactly the one dividing the discriminant -4 of $\mathbb{Z}[i]$. Ramification is where the two sheets of the cover collide, and it is detected algebraically by the discriminant—the same discriminant you met under the square root in the quadratic formula.

Exercise 1.1 ((easy) preimages of primes) — Let $\varphi: R \rightarrow S$ be a ring homomorphism and $\mathfrak{q} \subseteq S$ a prime ideal. Show $\varphi^{-1}(\mathfrak{q})$ is a prime ideal of R . (*Hint: $R/\varphi^{-1}(\mathfrak{q})$ injects into $S/\mathfrak{q}$.*) Then give an example showing the preimage of a *maximal* ideal need not be maximal.

1.3 The three pillars of the modern view: modules, localization, categories

What separates the modern treatment of this subject from its classical ancestor is which tools it treats as fundamental. Three of them carry almost all the weight, and naming them now—with

their motivation—turns the rest of these notes into a guided tour rather than a list of theorems.

Pillar 1: modules—linear algebra over a ring. You understand a group by its representations; you understand a ring by the things it acts on, its **modules**. A module is “a vector space whose scalars come from a ring instead of a field,” and almost every object we have met is one: ideals, quotient rings, and the ring itself are all R -modules; abelian groups are exactly $\mathbb{Z}$ -modules; vector spaces are exactly modules over a field. The modern slogan is that commutative algebra is the study of the category of R -modules. The single phenomenon that makes the theory deep is that, unlike vector spaces, *a module need not have a basis*—and measuring how badly bases fail (via free resolutions, Tor, Ext) is a large part of the subject. A vivid case to keep in mind: a module over $k[x]$ is a vector space equipped with a single linear operator (the action of x), so the classification of $k[x]$ -modules *is* the theory of Jordan and rational canonical forms.³ Linear algebra is already commutative algebra, in disguise.

Pillar 2: localization—geometry is local. To understand a space you look near each point; the algebraic counterpart is **localization**, the operation of inverting elements. Inverting everything outside a prime $\mathfrak{p}$ produces the **local ring** $R_{\mathfrak{p}}$, which sees only the behavior of R near the point $\mathfrak{p}$ —it is the ring of germs of functions there, exactly as in the theory of manifolds. The reward is the **local–global principle**: a great many properties hold for R if and only if they hold at every prime, so global theorems can be proved one point at a time. Localization is the technique that makes “study the space near a point” into algebra.

Pillar 3: categories—the connective language. We will constantly build new rings and modules out of old (quotients, products, tensor products, localizations, completions, free objects) and will want to say that each construction is *canonical*, pin it down *up to unique isomorphism*, and track how it interacts with maps. The precise language for all of this is category theory: **universal properties**, **functors**, and above all **adjoint functors**. It is not decoration. The fact that the tensor product is the left adjoint of Hom, and that “left adjoints are right exact,” organizes §§5–8 in a single stroke.⁴ Section 2 builds this language from nothing.

Equation 1.2 (The three pillars)

modules

localization

categories

Study a ring through what it acts on (modules); study it one point at a time (localization); and keep track of every construction by the maps it represents (categories).

1.4 If you remember one thing; and a map of the route ahead

Idea 1.2 (If you remember one thing)

A commutative ring is the ring of functions on a space, its spectrum $\text{Spec } R$, whose points are the prime ideals. Doing algebra to the ring *is* doing geometry to the space: ideals cut out

³The same slogan reaches much further: a module over a group ring $k[G]$ is precisely a representation of G , so representation theory is “the module theory of $k[G]$.” That is one of the doorways from commutative algebra toward the Langlands program, where representations are the central objects.

⁴This is the explicit thesis of Altman–Kleiman, who redo A–M with the categorical machinery in front: “Tensor Product is the left adjoint of Hom . . . any left adjoint preserves arbitrary direct limits” (A–K, Preface). We develop exactly this language, from scratch, in §2.

subspaces, quotients restrict to them, localization zooms in, and a ring map is a map of spaces running the other way. Every definition that follows is a tool for making this correspondence precise and computable—and, read in the other dialect, for turning arithmetic into the geometry of $\operatorname{Spec} \mathbb{Z}$.

With the thesis in hand, here is the shape of the journey, so you can see where each section fits in the arc.

Preview 1.2 (A map of the route ahead)

We first learn the *language* (§2: categories, universal properties, adjoints), then build the central *objects and the dictionary* (§3: $\operatorname{Spec} R$; §4: modules; §5: exact sequences). Next come the two everyday *tools*: localization (§6) and tensor products with flatness (§7), with their homological shadow $\operatorname{Tor}/\operatorname{Ext}$ (§8). We then do the *geometry of maps*—integral extensions and the Nullstellensatz (§§9–10)—and the *finiteness and decomposition* theory of Noetherian rings (§11). The climax is *dimension* (§12), refined locally by *completion* (§13). We cash out in dimension one—*DVRs and Dedekind domains*, the arithmetic payoff (§14)—and close by pointing to the *horizons*: schemes, sheaves, and the Langlands programs (§15).

Note 1.1 (Why a number theorist or a geometer should keep reading). Every object in these notes is load-bearing in modern research. Completions give the p -adic numbers $\mathbb{Z}_p$ and $\mathbb{Q}_p$; the *local fields* of the **local Langlands** correspondence are exactly these completions, and the *adeles* of **global Langlands** bundle all of them over $\operatorname{Spec} \mathbb{Z}$ at once. Dedekind domains and their class groups are the start of class field theory; regular local rings and dimension theory are the start of the geometry of singularities and **geometric Langlands**. We will flag these connections as the relevant tools come online—the present notes are, among other things, the commutative-algebra prerequisite for all of it.

1.5 Connections to further mathematics and physics

What you take from this section—a *ring is the functions on a space, and primes are its points*—is not a slogan you will outgrow; it is the organizing principle of several entire fields. Here is where it goes.

Within these notes. Every later section sharpens one corner of the dictionary: §3 makes $\operatorname{Spec} R$ and its topology precise; §4 reinterprets modules as “vector bundles” over $\operatorname{Spec} R$; §6 reads localization as zooming into an open set; §§9–10 turn ring extensions into maps of spaces and prove the Nullstellensatz; §12 measures the dimension we already glimpsed in the chains of primes. Keep the picture of §1 in view and the rest is commentary.

In mathematics. Gluing affine pieces $\operatorname{Spec} R$ produces **schemes** (Grothendieck), the objects of modern algebraic geometry (§15). On the arithmetic side, the §1.1 function-field analogy—number fields behaving like curves over $\mathbb{F}_q$ —is an engine of the Weil conjectures and the Langlands program; $\operatorname{Spec} \mathbb{Z}$ resembles a curve so closely that there is a developed dictionary, **arithmetic topology**, in which prime ideals play the role of *knots* and $\operatorname{Spec} \mathbb{Z}$ the role of a 3-manifold.⁵

⁵B. Mazur, *Remarks on the Alexander polynomial*; M. Morishita, *Knots and Primes*. Splitting of a prime mirrors the linking of knots.

In physics. The same duality—“a space is its commutative algebra of functions”—is exactly **Gelfand duality** (commutative C^* -algebras $\leftrightarrow$ compact Hausdorff spaces), which lets you recover a classical phase space from its algebra of observables; dropping commutativity is the starting point of Connes’s **noncommutative geometry**. Most strikingly for a field theorist:

Idea 1.3 (Spec of the chiral ring = the moduli space of vacua)

In a supersymmetric gauge theory the **chiral operators** form a *commutative* ring (the chiral ring), and the space of supersymmetric vacua is precisely its spectrum: the F -term equations cut out an ideal, gauge invariance imposes the quotient, and what remains is Spec of the resulting ring. The geometry you are learning is then physics—*irreducible components* are the distinct *branches* of vacua (Coulomb, Higgs, ...), the *singular points* where they meet are where extra states become massless, and the crossing lines $\text{Spec } k[x, y]/(xy)$ are the cartoon of two branches touching.^a

^aSee e.g. P. Argyres’s lectures on the moduli space of vacua; for D-branes probing $\text{Spec } R$ and the emergence of (noncommutative) geometry from gauge theory, Douglas–Moore and the quiver-gauge-theory literature.

D-branes make this literal: a brane probing spacetime sees the transverse geometry as the spectrum of its worldvolume chiral ring, and a stack of N branes promotes that ring to *matrices*, commutative geometry deforming into the noncommutative geometry of coincident branes. The one idea of §1 thus reaches from prime numbers to the vacua of quantum field theories.

1.6 Exercises

A first set, from the classic lists of Atiyah–Macdonald and Altman–Kleiman. These reward the big-picture viewpoint; do them with the dictionary of §1 in mind.

Exercise 1.2 ((easy) nilpotents and units (A–M Ch. 1, Ex. 1)) — Let x be nilpotent in a ring R . Show $1 + x$ is a unit (sum a finite geometric series), and deduce that *unit + nilpotent = unit*. (Geometrically, a function nonzero everywhere stays nonzero under an infinitesimal nudge; this fact powers Nakayama’s lemma in §4.)

Exercise 1.3 ((medium) the one-point spectrum (A–M Ch. 1, Ex. 10)) — Prove the equivalence, for a ring R , of: (i) R has exactly one prime ideal; (ii) every element is a unit or nilpotent; (iii) R modulo its nilpotent elements is a field. (These are the rings whose Spec is a single fat point—e.g. $k[x]/(x^n)$.)

Exercise 1.4 ((classic) the function-field analogy by hand) — Show $\mathbb{Z}$ and $k[x]$ are PIDs in which every nonzero element lies in only finitely many maximal ideals. For $k = \mathbb{F}_q$, compute $|k[x]/(f)|$ for f irreducible of degree d and compare with $|\mathbb{Z}/(p)| = p$. Conclude that $\mathbb{F}_q[x]$, not $\mathbb{C}[x]$, is the true arithmetic twin of $\mathbb{Z}$ —the residue fields are finite on both sides—as promised in §1.1.

Exercise 1.5 ((thinker) Artin builds an algebraic closure (A–M Ch. 1, Ex. 13)) — Let K be a field and form $A = K[x_f : f \text{ monic irreducible over } K]$, one variable per f . Let $\mathfrak{a} = (f(x_f))_f$. Show $\mathfrak{a} \neq (1)$, so (by the existence theorem of §3.2) it lies in a maximal ideal $\mathfrak{m}$; then $K_1 = A/\mathfrak{m}$ is an extension of K in which *every* f acquires a root. Iterate and take a union to build $\bar{K}$. (Marvel at what happened: a single maximal ideal manufactured roots for all polynomials at once—the abstract viewpoint doing thoroughly concrete work.)

Read alongside: A–M, Introduction and Ch. 1 (prime and maximal ideals, the nilradical, local rings); A–K, Preface (the categorical/adjoint thesis) and §§1–3; Aluffi, the prefaces to Ch. III for the “rings as the ground of geometry” framing. Skim these for orientation, not detail—the careful versions are §§3–4 below.

2 The categorical language, from scratch

The modern subject is written in the language of categories, so we learn just enough of it right away—introduced gently, with rings and modules as the running examples, and assuming no prior exposure. The payoff is enormous: a **universal property** lets us pin an object down by how it maps to everything else (no formulas), and **adjoint functors** organize half of what follows (tensor, localization, completion, free constructions) under a single slogan. We are not doing category theory for its own sake; we are learning the three or four ideas that the rest of the notes literally cannot be stated without.

2.1 Categories, functors, and natural transformations

We keep promising that a construction is “natural,” “canonical,” or that two objects are “the same kind of thing.” Category theory is the language that makes these words mean something. A word of reassurance before we start: we are not studying category theory for its own sake, and we need only its first few definitions. They earn their keep immediately, because the central constructions of commutative algebra—tensor products, localization, free modules—are cleanest, and are pinned down uniquely, by the categorical notions of *universal property* and *adjoint functor* introduced over the next pages.

Definition 2.1 (Category)

A **category** $\mathcal{C}$ consists of: a collection of **objects**; for each pair of objects A, B a set of **morphisms** (or **arrows**) $\text{Hom}_{\mathcal{C}}(A, B)$, written $f: A \rightarrow B$; a **composition** law $\text{Hom}_{\mathcal{C}}(B, C) \times \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{C}}(A, C)$, $(g, f) \mapsto g \circ f$; and for each object A an **identity** $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$. These must satisfy two axioms: composition is associative, $h \circ (g \circ f) = (h \circ g) \circ f$; and identities are neutral, $\text{id}_B \circ f = f = f \circ \text{id}_A$.

The definition is short because almost everything is an example. What it captures is “objects and the allowed transformations between them, with a sensible notion of doing one after another.”

Example 2.1 (The categories we will live in) • **Set**: sets and functions.

• **Grp, Ab**: groups (resp. abelian groups) and homomorphisms.

- **Ring**: commutative rings and ring homomorphisms (preserving 1). *This and the next are the homes of this course.*
- **R -Mod**: modules over a fixed ring R and R -linear maps. The central category of these notes; when $R = k$ is a field this is k -vector spaces, and when $R = \mathbb{Z}$ it is **Ab**.
- **Top**: topological spaces and continuous maps—the target of **Spec**.
- *A poset as a category.* Given a partially ordered set $(P, \leq)$, let the objects be the elements and put a single arrow $a \rightarrow b$ exactly when $a \leq b$. Composition is transitivity. For instance, the ideals of a ring R ordered by inclusion form a category. This example matters because it shows a “morphism” need not be a function at all—a category is more flexible than “sets with structure.”

Intuition 2.1 (It is all about the arrows)

The mental shift category theory asks for is this: stop looking *inside* an object (its elements, its construction) and look instead at the *arrows* into and out of it. We will see that an object is determined—up to unique isomorphism—by how it maps to and from everything else. That is the engine behind “universal properties” below, and it is why we can define the tensor product without ever choosing a basis.

Categories are related by functors, which are “maps of categories.”

Definition 2.2 (Functor)

A **functor** $F: \mathcal{C} \rightarrow \mathcal{D}$ assigns to each object A of $\mathcal{C}$ an object $F(A)$ of $\mathcal{D}$, and to each morphism $f: A \rightarrow B$ a morphism $F(f): F(A) \rightarrow F(B)$, preserving identities ($F(\text{id}_A) = \text{id}_{F(A)}$) and composition ($F(g \circ f) = F(g) \circ F(f)$). It is **covariant** as stated; a **contravariant** functor instead reverses arrows, $F(f): F(B) \rightarrow F(A)$, with $F(g \circ f) = F(f) \circ F(g)$.^a

^aEquivalently, a contravariant functor on $\mathcal{C}$ is a covariant functor on the *opposite category* $\mathcal{C}^{\text{op}}$, which has the same objects but all arrows reversed.

Functors are not exotic; the operations you already perform on rings and modules are mostly functors, and recognizing them as such is what lets us reason about them uniformly.

Example 2.2 (The operations of commutative algebra are functors) • *Forgetful functors*, e.g. $\text{Ring} \rightarrow \text{Set}$ (forget the ring structure) or $R\text{-Mod} \rightarrow \text{Ab}$ (forget scalar multiplication).

- *Free functors*, e.g. $\text{Set} \rightarrow R\text{-Mod}$ sending a set S to the free module $R^{(S)}$. (Soon: left adjoint to the forgetful functor.)
- *The spectrum* $\text{Spec}: \text{Ring} \rightarrow \text{Top}$, *contravariant*—a ring map $R \rightarrow S$ yields $\text{Spec } S \rightarrow \text{Spec } R$, as in §1.
- *The Hom functors.* For a fixed R -module N , $\text{Hom}_R(N, -)$ is a covariant functor $R\text{-Mod} \rightarrow R\text{-Mod}$, while $\text{Hom}_R(-, N)$ is contravariant.
- *Tensoring* $- \otimes_R N$ is a covariant functor (§7), and *localization* $S^{-1}(-)$ is a covariant functor

$R\text{-Mod} \rightarrow S^{-1}R\text{-Mod}$ (§6).

The whole later drama—left versus right exactness, Tor and Ext—is about how these last functors fail to preserve exact sequences. That is why we set them up as functors now.

Finally, functors themselves can be compared, and the comparison is where the word “natural” acquires its technical meaning.

Definition 2.3 (Natural transformation)

Given two functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$, a **natural transformation** $\eta: F \Rightarrow G$ is a choice, for every object A , of a morphism $\eta_A: F(A) \rightarrow G(A)$, such that for every morphism $f: A \rightarrow B$ the square

$$\begin{array}{ccc} F(A) & \xrightarrow{\eta_A} & G(A) \\ F(f) \downarrow & & \downarrow G(f) \\ F(B) & \xrightarrow{\eta_B} & G(B) \end{array}$$

commutes. If every η_A is an isomorphism, η is a **natural isomorphism** and we write $F \cong G$.

Intuition 2.2 (“Natural” means “uniformly, without choices”)

A natural transformation is the same construction carried out at every object, compatibly with all morphisms. This is the rigorous meaning of “canonical.” The textbook contrast is from linear algebra: for finite-dimensional vector spaces the map $V \rightarrow V^{**}$ into the double dual is natural (it needs no basis and commutes with every linear map), whereas the map $V \rightarrow V^*$ to the dual is *not* natural—you can only build one by choosing a basis, and different choices give incompatible maps. “Canonical” was always trying to mean “natural”; now it does. A purely algebraic instance, which you will meet in §6: for a multiplicative set $S \subseteq R$ the localization maps $M \rightarrow S^{-1}M$, $m \mapsto m/1$, are natural in M —the same construction at every module, compatible with every R -linear map—so localization comes packaged with a canonical natural transformation $\text{id} \Rightarrow S^{-1}(-)$ from the identity functor.

Exercise 2.1 ((easy) Hom is a functor) — Fix an R -module N . Check that $\text{Hom}_R(N, -)$ is a covariant functor: describe what it does to a morphism $g: M \rightarrow M'$, and verify it preserves identities and composition. Then do the same for the contravariant $\text{Hom}_R(-, N)$.

Exercise 2.2 ((classic) natural versus chosen) — On finite-dimensional k -vector spaces, show $V \mapsto V^* = \text{Hom}_k(V, k)$ is a contravariant functor and $V \mapsto V^{**}$ is a covariant functor, and that the evaluation maps $\text{ev}_V: V \rightarrow V^{**}$ assemble into a natural isomorphism. Explain in one sentence why there is no natural isomorphism $V \cong V^*$. (*This is the cleanest possible illustration of the definition.*)

2.2 Universal properties: characterizing an object by its maps

Here is the single most useful technique category theory gives us. A **universal property** characterizes an object by the morphisms into or out of it—never by its internal construction. The reason to care

is twofold: a universal property defines a construction *without arbitrary choices*, and it automatically proves the construction is *unique up to a unique isomorphism*. After we state a universal property we will always also give an explicit model (a concrete construction) and check the two describe the same object; the universal property tells us *what* the thing is, the model proves it *exists*.

The simplest universal properties name an object by a single mapping condition.

Definition 2.4 (Initial and terminal objects)

An object I of $\mathbf{C}$ is **initial** if for every object A there is exactly one morphism $I \rightarrow A$. It is **terminal** (or **final**) if for every A there is exactly one morphism $A \rightarrow I$. An object that is both is a **zero object**.

Example 2.3 (Initial and terminal, in our categories)

In **Set**: the empty set $\emptyset$ is initial, any one-point set is terminal. In **R -Mod**: the zero module is both—a zero object. In **Ring**: the *zero ring* is terminal (every ring maps onto it), and—crucially— $\mathbb{Z}$ is *initial*, because every ring R receives exactly one ring homomorphism from $\mathbb{Z}$ (it must send $1 \mapsto 1$, hence $n \mapsto n \cdot 1_R$, and this is forced). We return to how special this makes $\mathbb{Z}$ in §2.3.

The reason universal properties are trustworthy is the following two-line theorem, which we prove because it is the engine that makes “*the* tensor product,” “*the* localization,” etc. well-defined.

Theorem 2.1 (Universal properties pin objects down)

Any two initial objects of a category are isomorphic, by a *unique* isomorphism. (Dually for terminal objects.)

Proof. Let I, I' be initial. Initiality of I gives a unique map $u: I \rightarrow I'$; initiality of I' gives a unique $v: I' \rightarrow I$. Then $v \circ u: I \rightarrow I$ is a map from I to itself, but initiality of I says there is only *one* such map, and id_I is one, so $v \circ u = \text{id}_I$. Symmetrically $u \circ v = \text{id}_{I'}$. Hence u is an isomorphism, and it was the unique map $I \rightarrow I'$ to begin with.

Technique 2.1 (How to read and use a universal property)

Every universal property in this book is secretly the statement “*this object is initial (or terminal) in an auxiliary category of solutions to a mapping problem.*” So it always buys you the same three things: (i) a recipe—“a map out of X is the same as the following data”; (ii) uniqueness up to unique isomorphism, for free, by the theorem above; and (iii) a way to *construct* maps out of (or into) X without touching its internals. Don’t ask what the object *is*; ask what maps to and from it *do*.

We now meet the workhorse universal properties. In each, notice the rhythm: *universal property*, then *concrete model*, then *why they agree*.

Example 2.4 (Quotient ring R/I : the universal way to kill I)

Universal property. The quotient map $\pi: R \rightarrow R/I$ is the universal ring homomorphism out of R that sends I to 0: for every ring map $\varphi: R \rightarrow S$ with $\varphi(I) = 0$ there is a *unique* $\bar{\varphi}: R/I \rightarrow S$ with $\bar{\varphi} \circ \pi = \varphi$. *Concrete model.* Cosets $r + I$, with the inherited operations. *Why they agree.* The map $\bar{\varphi}(r + I) \stackrel{\text{def}}{=} \varphi(r)$ is well-defined exactly because $\varphi(I) = 0$, and it is the only possible value—this is the first isomorphism theorem, read as a universal property.^a

^aAluffi, Ch. III §3, develops quotient rings precisely this way; the same pattern (a quotient as a universal “map that kills a subobject”) recurs for modules in §4 below.

Example 2.5 (Free module $R^{(S)}$: maps out of it are just functions on S)

Universal property. For a set S there is a module $R^{(S)}$ with an inclusion $S \hookrightarrow R^{(S)}$ such that every set-function $S \rightarrow M$ into an R -module M extends *uniquely* to an R -linear map $R^{(S)} \rightarrow M$. *Concrete model.* Finite formal combinations $\sum_s r_s e_s$ ($r_s \in R$, almost all 0). *Why they agree.* “Define a linear map by saying where the generators go, then extend linearly” is exactly the unique extension—this is the basis-free phrasing of the most familiar move in linear algebra.

Example 2.6 (Polynomial ring $R[x]$: “plugging in” is a universal property)

Universal property. $R[x]$ is the free commutative R -algebra on one generator: for any R -algebra S , an R -algebra map $R[x] \rightarrow S$ is the *same data* as a choice of one element of S (the image of x). *Concrete model.* Polynomials $\sum a_i x^i$. *Why they agree.* Sending $x \mapsto s$ and extending by the algebra laws is the substitution (“evaluation”) homomorphism, and it is the unique extension—this is precisely *why* you are allowed to plug anything you like into a polynomial.^a

^aAluffi, Ch. III §2.2, “Universal property of polynomial rings.” This example is the prototype for the *functor of points* we mention in the note below.

Note 2.1 (A first glimpse of the functor of points). The polynomial example says $\text{Hom}_{\text{Ring}}(R[x], S) \cong S$ *naturally in* S : the ring $R[x]$ “is” the rule that assigns to each ring S its underlying set of elements—the *affine line*. This is Grothendieck’s **functor of points** idea: an affine scheme $\text{Spec } R$ is encoded by the functor $S \mapsto \text{Hom}_{\text{Ring}}(R, S)$ of its “ S -valued points.” We are not using this yet, but it is the reason the categorical language is not optional in modern algebraic geometry (§15).

Exercise 2.3 ((easy) the two ends of Ring) — Prove carefully that $\mathbb{Z}$ is initial and the zero ring is terminal in **Ring**. Where exactly does “ring homomorphisms preserve 1” get used?

Exercise 2.4 ((classic) the quotient as a universal property) — State the universal property of R/I precisely and prove it from the coset construction. Then deduce the first isomorphism theorem (any $\varphi: R \rightarrow S$ factors as $R \twoheadrightarrow R/\ker \varphi \xrightarrow{\sim} \text{im } \varphi \hookrightarrow S$) from it.

2.3 Limits, colimits, and the initial/terminal objects (why $\mathbb{Z}$ is special)

Initial and terminal objects are the smallest universal properties; products, quotients, intersections, and unions are all instances of two general patterns called **limits** and **colimits**. We introduce them lightly—enough to recognize the constructions of commutative algebra as members of these two families, because that recognition is what makes the adjoint-functor theorems of §2.4 so powerful.

Definition 2.5 (Limit and colimit, informally)

A **diagram** in $\mathcal{C}$ is a collection of objects and morphisms (a picture of arrows). A **limit** of the diagram is a universal object mapping *to* all of it: an object L with compatible maps to every object of the diagram, through which any other such “compatible family of maps” factors uniquely. A **colimit** is the dual: a universal object receiving maps *from* all of it. (Limits generalize “intersection/product”; colimits generalize “union/glue.”)

Example 2.7 (The (co)limits you already use) • Products and coproducts. The **product**

$\prod A_i$ is the limit of a diagram with no arrows (it has projections to each A_i); the **coproduct** is the colimit (it has inclusions from each A_i). In $R\text{-Mod}$ a finite product is the direct sum $\bigoplus A_i$, which is *also* the coproduct—a biproduct. In Ring the product is the usual $R \times S$, while the coproduct is the tensor product $R \otimes_{\mathbb{Z}} S$ (§7).

- *Kernels and cokernels.* In $R\text{-Mod}$, $\ker f$ is a limit and $\text{coker } f$ is a colimit; quotient modules are colimits.
- *Direct limits* $\varinjlim$, colimits over a directed system, are pervasive in commutative algebra: every module is the direct limit of its finitely generated submodules, and (§6) the localization $R_{\mathfrak{p}} = \varinjlim_{f \notin \mathfrak{p}} R_f$ is a direct limit.
- *Inverse limits* $\varprojlim$ are the limits we complete with: the p -adic integers are $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$, and more generally $\widehat{R} = \varprojlim_n R/I^n$ (§13).

Not every category has all such (co)limits on hand—the category of fields, for instance, has no products, since a product $k \times k'$ of two fields is not a field (it has zero-divisors). What makes $R\text{-Mod}$ and Ring so pleasant to work in is precisely that they *are* complete and cocomplete: every diagram in them has both a limit and a colimit. We will use this silently and often.

Now we can say precisely why $\mathbb{Z}$ deserved a starring role in §1, and why number theory is, structurally, “geometry over a single base point.”

Idea 2.1 (Why $\mathbb{Z}$ is special, and why everything lives over $\text{Spec } \mathbb{Z}$)

$\mathbb{Z}$ is the *initial* object of Ring : every ring receives a unique ring map from $\mathbb{Z}$. Translating through the contravariant dictionary of §1, this says $\text{Spec } \mathbb{Z}$ is the *terminal* object among affine schemes—*every* commutative ring, hence every affine scheme, has a unique map to $\text{Spec } \mathbb{Z}$. So all of commutative algebra and algebraic geometry takes place “over $\text{Spec } \mathbb{Z}$,” and the special case of rings that happen to be algebras over a field k is the case of geometry over the point $\text{Spec } k$. Number theory is then exactly the geometry of $\text{Spec } \mathbb{Z}$ and its covers (§1’s $\mathbb{Z}[i]$ example); this “arithmetic = geometry over the terminal base $\text{Spec } \mathbb{Z}$ ” is the conceptual seed of Arakelov

theory and much of modern arithmetic geometry.

Exercise 2.5 ((easy) products and coproducts by hand) — Identify the product and the coproduct of two objects in each of **Set**, **Ab**, and **Ring**. (You should find disjoint union vs. cartesian product in **Set**; direct sum = direct product for two objects in **Ab**; and $R \times S$ vs. $R \otimes_{\mathbb{Z}} S$ in **Ring**.)

Exercise 2.6 ((medium) completion and localization as (co)limits) — Exhibit the structure maps making $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ an inverse limit, and (looking ahead to §6) describe the directed system whose direct limit is R_p . In each case, what is the universal property doing for you?

2.4 Adjoint functors: the single most important idea in the book

Everything so far converges here. **Adjoint functors** are the organizing principle of the subject: tensor products, localization, free modules, base change, and completion are all one half of an adjunction, and two short theorems about adjunctions will let us predict their behavior toward exact sequences before we compute anything. We motivate the definition with the example you have already seen twice.

Recall the free module: a set-function $S \rightarrow U(M)$ (where U forgets the module structure) is the same data as an R -linear map $R^{(S)} \rightarrow M$. Spelled out, there is a bijection, natural in both S and M ,

$$\mathrm{Hom}_{R\text{-Mod}}(R^{(S)}, M) \cong \mathrm{Hom}_{\mathrm{Set}}(S, U(M)).$$

This bijection is an adjunction between the free functor $F: S \mapsto R^{(S)}$ and the forgetful functor U .

Definition 2.6 (Adjoint functors)

Functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{C}$ form an **adjunction**, with F **left adjoint** to G (and G **right adjoint** to F), written $F \dashv G$, if there is a bijection

$$\mathrm{Hom}_{\mathcal{D}}(F(c), d) \cong \mathrm{Hom}_{\mathcal{C}}(c, G(d))$$

natural in $c \in \mathcal{C}$ and $d \in \mathcal{D}$.

Intuition 2.3 (Left adjoints build, right adjoints forget)

Picture G as “forgetting” some structure and F as “freely building it back.” The adjunction says: to map a freely-built object $F(c)$ into d is no more and no less than to map the raw material c into the forgotten shadow $G(d)$ —you specify the map on generators and the structure forces the rest. Left adjoints are the *constructive, generators-and-relations* functors (free modules, tensor, localization, base change); right adjoints are the *forgetful, limit-like* functors (the forgetful functor, Hom). Keeping this picture straight tells you which side of every construction you are on.

The reason this single definition reorganizes the subject is the following list, every entry of which we will use.

Example 2.8 (The adjunctions that run commutative algebra) • *Free* $\dashv$ *forgetful* (the model): $R^{(-)} \dashv U$ for modules, and $\mathbb{Z}[-] \dashv U$ for rings (the polynomial ring $\mathbb{Z}[S]$ is the free commutative ring on a set S).

- *Tensor* $\dashv$ *Hom* (“adjoint associativity”), the most-used adjunction in the book:

$$\mathrm{Hom}_R(M \otimes_R N, P) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P)),$$

i.e. $- \otimes_R N \dashv \mathrm{Hom}_R(N, -)$ (§7).

- *Base change* $\dashv$ *restriction of scalars*: for a ring map $R \rightarrow S$, the functor $S \otimes_R -$ is left adjoint to restricting an S -module to an R -module. (§7.)
- *Localization* $\dashv$ *restriction*: $S^{-1}(-)$ is left adjoint to restriction along $R \rightarrow S^{-1}R$ —localization is the special case of base change with $S^{-1}R$ (§6).

What makes these worth naming all at once is that adjointness, by itself, predicts how each functor treats limits and colimits—and hence exact sequences.

Theorem 2.2 (The slogan that organizes §§5–8)

Left adjoints preserve colimits (in particular cokernels, direct sums, and direct limits);^a right adjoints preserve limits (in particular kernels, products, and inverse limits). Consequently $- \otimes_R N$ is right exact and $\mathrm{Hom}_R(N, -)$, $\mathrm{Hom}_R(-, N)$ are left exact.^b Moreover adjoints, when they exist, are unique up to natural isomorphism—so “*the*” tensor product and “*the*” localization are well-defined.

^aThe proof is a one-line calculation in the spirit of the “it is all about the arrows” box. If $F \dashv G$ and $c = \varinjlim c_i$, then for every object d we have $\mathrm{Hom}(F(\varinjlim c_i), d) \cong \mathrm{Hom}(\varinjlim c_i, Gd) \cong \varprojlim \mathrm{Hom}(c_i, Gd) \cong \varprojlim \mathrm{Hom}(Fc_i, d) \cong \mathrm{Hom}(\varinjlim Fc_i, d)$, naturally in d (the middle step is that $\mathrm{Hom}(-, Gd)$ turns colimits into limits). Since an object is pinned down by the maps out of it—the **Yoneda lemma**—this forces $F(\varinjlim c_i) \cong \varinjlim Fc_i$. The statement for right adjoints is the dual.

^bThis is exactly the structural observation A–K put at the center of their treatment: from “Tensor is the left adjoint of Hom” and “a left adjoint preserves direct limits/colimits” they deduce right-exactness of tensor and much else (A–K, Preface and §§6–9). A–M prove the same isomorphism (their “adjoint associativity”) but, writing before this language was standard, do not name it an adjunction; we do.

Equation 2.1 (Carry this for the next four sections)

left adjoint $\Rightarrow$ right exact (preserves colimits),

right adjoint $\Rightarrow$ left exact (preserves limits).

Tensor and localization are left adjoints; Hom is a right adjoint. The *failure* of the missing half—how far $\otimes$ is from left exact, how far Hom is from right exact—is measured by the derived functors Tor and Ext (§8). Half of homological algebra is bookkeeping for this one slogan.

Note 2.2 (The same idea, all the way up to Langlands). Adjunctions are not special to commutative algebra; they are how mathematics expresses “free/forgetful” and “induction/restriction” everywhere. In representation theory, induction and restriction of representations are adjoint (Frobenius reciprocity); in the **Langlands** program their analogues—parabolic induction and Jacquet restriction—are adjoint functors, and the geometric Langlands correspondence is, at bottom, an equivalence between two (derived) categories implemented by functors of exactly this kind. The language you are learning here is the same language those theories are written in.

Exercise 2.7 ((easy) the free module adjunction) — Write out the natural bijection

$$\mathrm{Hom}_{R\text{-Mod}}(R^{(S)}, M) \cong \mathrm{Hom}_{\mathrm{Set}}(S, U(M))$$

explicitly in both directions, and check it is natural in M . Conclude $R^{(-)} \dashv U$.

Exercise 2.8 ((classic) adjoint associativity) — Construct natural maps in both directions between $\mathrm{Hom}_R(M \otimes_R N, P)$ and $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$ and show they are mutually inverse. (You may take the generators-and-relations construction of $\otimes$ as known from §7; the point is to see the adjunction concretely.)

Exercise 2.9 ((medium) right-exactness for free) — Granting the theorem that left adjoints preserve cokernels, deduce that $- \otimes_R N$ is right exact: applying it to $M' \rightarrow M \rightarrow M'' \rightarrow 0$ gives an exact $M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0$. Where, precisely, did “left adjoint” do the work?

Exercise 2.10 ((thinker) adjoints are essentially unique) — Show that if F has two right adjoints G, G' then $G \cong G'$ naturally. (Hint: both represent the same functor $d \mapsto \mathrm{Hom}_D(F(-), d)$; use the uniqueness theorem of §2.2.) Conclude that “the” tensor product and “the” localization are well-defined.

2.5 Connections to further mathematics and physics

The categorical vocabulary of this section is the lingua franca of modern mathematics and of a surprising amount of modern physics; here is a map of where it carries you.

Within these notes and mathematics. Universal properties will *define* every construction we build—tensor products, localizations, completions—and adjointness will predict their exactness for free (§§5–8). One level up, repairing the failure of functors to be exact is **homological algebra**, which lives most naturally in **derived** and **triangulated categories**; the functor-of-points idea (§2.2) is the gateway to **moduli spaces** and **stacks**; and sheaves and topoi rebuild geometry on categorical foundations. Learning to think in arrows now pays compound interest.

In physics. Category theory is not ornamental in modern theoretical physics—in several places it *is* the definition.

Idea 2.2 (A TQFT is a functor)

The Atiyah–Segal axioms *define* a topological quantum field theory as a symmetric monoidal functor $Z: \text{Bord}_n \rightarrow \text{Vect}$ from a cobordism category (manifolds as objects, the spacetimes between them as morphisms) to vector spaces: gluing spacetimes is composing morphisms, and the partition function is what the functor does to a closed manifold. The whole modern theory of **generalized symmetries**—symmetry operators as topological defects that *fuse*, forming a (higher) **fusion category**—is this categorical language taken seriously, and it is exactly where your gauge-theory notes were heading.^a

^aM. Atiyah, *Topological quantum field theories* (1988); J. Lurie’s cobordism-hypothesis classification of fully extended TQFTs via higher categories; and the generalized-symmetries literature following Gaiotto–Kapustin–Seiberg–Willett.

Adjunctions and equivalences are dualities. Many physical dualities are, precisely, equivalences of categories. The **derived category** of coherent sheaves $D^b(\text{Coh } X)$ is the category of B-type D-branes, and Kontsevich’s **homological mirror symmetry** posits an equivalence between it and the Fukaya category of the mirror—mirror symmetry *is* a categorical equivalence. And Kapustin–Witten realize the **geometric Langlands** correspondence as the statement that two categories attached to a Riemann surface, built from S -dual $\mathcal{N} = 4$ super Yang–Mills theories, are equivalent. The functors and adjoints of §2 are the grammar in which these dualities are written.

2.6 Exercises

In Atiyah–Macdonald and Altman–Kleiman the categorical exercises live inside the later chapters where the constructions appear (tensor, localization, limits)—we mine those in §§4–7. For the pure categorical content here we follow Aluffi’s Chapter I; the last two problems preview the adjunctions A–K builds on.

Exercise 2.11 ((easy) initial and terminal, everywhere) — Identify the initial and terminal objects (when they exist) in **Set**, **Grp**, **Ring**, **R -Mod**, and the category of a poset P . Where is there a zero object? (*Recall $\mathbb{Z}$ is initial in **Ring**; what does that say about $\text{Spec } \mathbb{Z}$?*)

Exercise 2.12 ((classic) an epimorphism that is not surjective (Aluffi Ch. I)) — Show the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is an *epimorphism* in **Ring**: any two ring maps $\mathbb{Q} \rightarrow S$ that agree on $\mathbb{Z}$ are equal. Conclude that “epimorphism” and “surjection” genuinely diverge—an early lesson that categories see arrows, not elements.

Exercise 2.13 ((medium) the coproduct of rings is the tensor product) — Granting the construction of $R \otimes_{\mathbb{Z}} S$ (§7), show that with the maps $r \mapsto r \otimes 1$ and $s \mapsto 1 \otimes s$ it satisfies the universal property of the *coproduct* in **Ring**: a pair of maps $R \rightarrow T$, $S \rightarrow T$ corresponds to a unique map $R \otimes_{\mathbb{Z}} S \rightarrow T$. (*Geometrically this is the product of spaces $\text{Spec } R \times_{\text{Spec } \mathbb{Z}} \text{Spec } S$ —product of spaces = tensor product of rings.*)

Exercise 2.14 ((medium) free $\dashv$ forgetful for rings) — Let $U: \mathbf{Ring} \rightarrow \mathbf{Set}$ be the forgetful functor. Show its left adjoint sends a set S to the polynomial ring $\mathbb{Z}[S]$: that is, $\mathrm{Hom}_{\mathbf{Ring}}(\mathbb{Z}[S], T) \cong \mathrm{Hom}_{\mathbf{Set}}(S, U(T))$ naturally. Which universal property of §2.2 is this?

Exercise 2.15 ((thinker) Yoneda in miniature) — Show that if two objects X, Y of a category represent the same functor— $\mathrm{Hom}(X, -) \cong \mathrm{Hom}(Y, -)$ naturally—then $X \cong Y$ by a canonical isomorphism. (*This is the precise sense in which an object is determined by the maps out of it: the principle behind every universal property, and behind the proof in §2.4 that left adjoints preserve colimits.*)

Read alongside: Aluffi Ch. I §§3–5 (categories, morphisms, universal properties; initial/final objects, products, coproducts) is the gentle, example-rich source for §§2.1–2.3; Ch. VIII §1 covers functors and natural transformations, and §2 the tensor/Tor adjunction. A–K threads adjoints through the whole book—its Preface and §§6–9 are the place to watch “left adjoint $\Rightarrow$ right exact” do real work. For category theory in its own right, S. Mac Lane, *Categories for the Working Mathematician*, is the standard reference.

3 Rings, ideals, and the spectrum

Now the geometry begins. We review rings, ideals, and quotients quickly—but with a modern slant—and then build the central object: the **prime spectrum** $\mathrm{Spec} R$, topologized by the **Zariski topology**. The conceptual leap, and the one worth slowing down for, is *why the points are the prime ideals* (not just the maximal ones) and how the elements of R act as functions on $\mathrm{Spec} R$. We make Spec into a contravariant functor so that algebra and geometry translate automatically, and we draw the first pictures.

3.1 Rings, homomorphisms, ideals, and quotients: a fast modern review

This material you have met before, so we move quickly—but with the geometric reading turned on, because that reading is the point. Throughout, **ring** means *commutative ring with 1*, and a **ring homomorphism** preserves 1 (so that $\mathbb{Z}$ stays initial, §2).

Definition 3.1 (Ideal, quotient ring)

An **ideal** $I \subseteq R$ is an additive subgroup that absorbs multiplication: $r \in R, a \in I \Rightarrow ra \in I$. The **quotient ring** R/I has elements the cosets $r + I$ with the inherited operations, and comes with the quotient map $\pi: R \rightarrow R/I$.

Why single out ideals among all subsets? Because *ideals are exactly the kernels of ring homomorphisms*—the subsets you can quotient by. If $\varphi: R \rightarrow S$ is a homomorphism then $\ker \varphi = \varphi^{-1}(0)$ is an ideal, and conversely every ideal I is $\ker(\pi: R \rightarrow R/I)$. So the slogan “ideal = kernel = quotient” is a chain of synonyms, and geometrically (recall §1) a quotient $R \rightarrow R/I$ will be the restriction of functions to the closed subspace cut out by “ $I = 0$.” Ideals are how we impose equations.

There are a handful of standard *operations* on ideals, each with a geometric meaning expressed through the **vanishing locus** $V(I) \stackrel{\mathrm{def}}{=} \{\mathfrak{p} \in \mathrm{Spec} R : I \subseteq \mathfrak{p}\}$ —the set of points where every element of I vanishes (we topologize it in §3.3). Given ideals I, J :

- the **sum** $I + J = \{a + b : a \in I, b \in J\}$, the smallest ideal containing both (geometrically: imposing *both* sets of equations, so $V(I + J) = V(I) \cap V(J)$);
- the **product** $IJ = (\{ab : a \in I, b \in J\})$, the ideal generated by products;
- the **intersection** $I \cap J$;
- the ideal (E) **generated** by a subset $E \subseteq R$, the smallest ideal containing E (all finite R -combinations of elements of E).

We always have $IJ \subseteq I \cap J$. The single most useful structural fact is the correspondence between a quotient and the part of R above I .

Proposition 3.1 (Correspondence theorem)

The map $\bar{J} \mapsto \pi^{-1}(\bar{J})$ is an inclusion-preserving bijection

$$\{\text{ideals of } R/I\} \xrightarrow{\sim} \{\text{ideals of } R \text{ that contain } I\},$$

and it respects sums, intersections, products, primeness, and maximality.

Proof. The quotient map π is surjective, so $\bar{J} \mapsto \pi^{-1}(\bar{J})$ and $K \mapsto \pi(K) = K/I$ are mutually inverse on the displayed sets (an ideal of R pulls back from R/I iff it contains $I = \ker \pi$). Both maps preserve inclusions, and the listed operations are preserved because π is a surjective homomorphism. The clauses about prime and maximal ideals (defined in §3.2) follow from the third isomorphism theorem: for $J \supseteq I$ one has $R/J \cong (R/I)/(J/I)$, so J is prime (resp. maximal) in R exactly when J/I is prime (resp. maximal) in R/I .

Read geometrically, the correspondence theorem says: *the closed subspace $V(I)$ has its own geometry, and that geometry is the part of R 's geometry lying over I .* Passing to R/I throws away everything off $V(I)$ and keeps exactly what sits on it.

Example 3.1 (The ideals you already know)

In $\mathbb{Z}$ every ideal is (n) for a unique $n \geq 0$ (it is a **principal ideal domain**, PID). In $k[x]$ every ideal is (f) (also a PID, by division with remainder). In $k[x, y]$ *not* every ideal is principal: (x, y) —polynomials with zero constant term—needs two generators, which is the algebra of the fact that a point of the plane is cut out by two equations. In the zero ring 0 (where $1 = 0$) the only ideal is everything; its Spec will be empty.

Exercise 3.1 ((easy) the correspondence in action) — List all ideals of $\mathbb{Z}/12\mathbb{Z}$ using the correspondence theorem, and match each to an ideal of $\mathbb{Z}$ containing (12) . Which are prime? (You will be able to answer the last part after §3.2; guess now and check later.)

3.2 Prime and maximal ideals, and why the primes are the points

This is the conceptual hinge of the whole subject, so we go slowly. We already decided in §1 that the points of $\text{Spec } R$ should be the prime ideals; now we define them carefully, gather many examples, and earn the slogan.

Definition 3.2 (Prime and maximal ideals)

An ideal $\mathfrak{p} \subsetneq R$ is **prime** if $ab \in \mathfrak{p} \Rightarrow a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. An ideal $\mathfrak{m} \subsetneq R$ is **maximal** if no ideal lies strictly between $\mathfrak{m}$ and R . (In both, the condition $\subsetneq R$, i.e. $1 \notin \mathfrak{p}$, excludes the whole ring.)

The definitions look unrelated until you pass to quotients, where they become transparent.

Proposition 3.2 (Primes and maximals via quotients)

For an ideal $I \subsetneq R$:

$$I \text{ is prime} \iff R/I \text{ is an integral domain;} \quad I \text{ is maximal} \iff R/I \text{ is a field.}$$

In particular every maximal ideal is prime (a field is a domain).

Proof. R/I has no zero-divisors $\iff (\bar{a}\bar{b} = 0 \Rightarrow \bar{a} = 0 \text{ or } \bar{b} = 0) \iff (ab \in I \Rightarrow a \in I \text{ or } b \in I)$, which is primeness. For the second: by the correspondence theorem R/I has *no* ideals other than 0 and itself $\iff I$ is maximal, and a nonzero commutative ring is a field $\iff$ its only ideals are 0 and the whole ring (if $x \neq 0$ then $(x) = R$, so x is a unit).⁶ Since fields are domains, maximal $\Rightarrow$ prime.

Intuition 3.1 (Domain = irreducible, field = point)

Recall the reading of §1: R/I is the ring of functions on the subspace $V(I)$. A *domain* has no zero-divisors, i.e. no two nonzero functions whose product vanishes identically—so $V(\mathfrak{p})$ cannot be broken into two pieces; it is **irreducible**. A *field* is the ring of functions on a single reduced point (you can divide by anything nonzero, because nothing nonzero vanishes). So:

$$\text{prime} \longleftrightarrow \text{irreducible closed piece,} \quad \text{maximal} \longleftrightarrow \text{a (classical) point.}$$

A maximal ideal is the smallest, most “point-like” kind of prime; a non-maximal prime is a larger irreducible piece (a curve, a surface) seen as a single **generic point**.

Example 3.2 (Primes and maximals across our running rings) • $\mathbb{Z}$: primes are (0) and (p) for p a prime number; the maximal ones are exactly the (p) (since $\mathbb{Z}/(p) = \mathbb{F}_p$ is a field, while $\mathbb{Z}/(0) = \mathbb{Z}$ is a domain but not a field). So (0) is prime-but-not-maximal: the generic point.

- $k[x]$: primes are (0) and (f) with f irreducible; maximal are the (f) . For $k = \bar{k}$ these are $(x - a)$, the points $a \in k$.
- $k[x, y]$: three tiers— (0) (prime, the whole plane); (f) with f irreducible (prime, a curve); and (for $k = \bar{k}$) $(x - a, y - b)$ (maximal, a point). The chain $(0) \subsetneq (f) \subsetneq (x - a, y - b)$ has length two—a preview of $\dim = 2$ (§12).

⁶A–M Ch. 1, Props. 1.2–1.4; A–K §2. The quotient characterization is the one to internalize—it converts every question about primes/maximals into a question about domains/fields.

- **A field k :** the only ideal $\subsetneq k$ is (0) , which is maximal; $\text{Spec } k$ is a single point.
- **Non-examples:** $(4) \subseteq \mathbb{Z}$ is not prime ($2 \cdot 2 \in (4)$ but $2 \notin (4)$); $(xy) \subseteq k[x, y]$ is not prime ($x \cdot y \in (xy)$, neither factor in). Each “factors,” which is exactly reducibility: $V(xy)$ is the union of the two axes.

Two facts make primes and maximals usable. First, maximal ideals always exist—so $\text{Spec } R$ is never empty for $R \neq 0$.

Theorem 3.1 (Existence of maximal ideals)

Every ring $R \neq 0$ has a maximal ideal, and more generally every proper ideal $I \subsetneq R$ is contained in a maximal ideal.

Proof. Apply Zorn’s lemma to the set of proper ideals containing I , ordered by inclusion: the union of a chain of proper ideals is a proper ideal (it still omits 1), so chains have upper bounds, and a maximal element is a maximal ideal.⁷

Second, some rings have just *one* maximal ideal, and these turn out to be the natural “local” objects of geometry.

Definition 3.3 (Local ring)

A **local ring** is a ring with exactly one maximal ideal $\mathfrak{m}$. Its **residue field** is $\kappa = R/\mathfrak{m}$.

Example 3.3 (Local rings to keep in mind)

A field is local ($\mathfrak{m} = (0)$). The formal power series ring $k[[x]]$ is local, with maximal ideal (x) and residue field k : a power series is a unit iff its constant term is nonzero, so the non-units are exactly (x) .^a A number-theoretic cousin is $\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}$, local with maximal ideal (p) and residue field $\mathbb{F}_p$ —the arithmetic “germ” of $\mathbb{Z}$ at the prime p . We will manufacture local rings at will by **localization** in §6—indeed $R_{\mathfrak{p}}$, the ring R “seen near the point $\mathfrak{p}$,” is local with maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$, which is the entire reason the construction exists.

^aIn any ring, if the non-units form an ideal $\mathfrak{m}$ then R is local with maximal ideal $\mathfrak{m}$ (A–M Ch. 1, Prop. 1.6). “Local” is the algebraic form of “germ at a point.”

Note 3.1 (The two meanings of “prime,” reconciled). In a ring of integers $\mathcal{O}_K$ the nonzero prime ideals are the genuine “primes” of number theory—and in $\mathbb{Z}$ itself the prime ideal (p) is literally the prime number p . The extra point (0) (present whenever R is a domain) is the generic point, the home of the fraction field $\text{Frac}(R)$. So “prime ideal” simultaneously generalizes “prime number” (the maximal ones) and “irreducible subvariety” (all of them); this double life is precisely why the same theorems serve number theory and geometry.

⁷A–M Ch. 1, Thm. 1.3 and Cor. 1.4. The only subtle point is that the union of a chain misses 1 because each member does; this is where “proper” ($1 \notin I$) does the work.

Exercise 3.2 ((easy) domains and fields by quotient) — Verify directly that $(x) \subseteq k[x, y]$ is prime but not maximal by computing $k[x, y]/(x)$, and that (x, y) is maximal by computing $k[x, y]/(x, y)$.

Exercise 3.3 ((medium) every ring has a prime) — Deduce from the existence theorem that $\text{Spec } R \neq \emptyset$ for $R \neq 0$. Then show the set of prime ideals has *minimal* elements under inclusion (the **minimal primes**). (*Hint: Zorn, downward, using that an intersection of a chain of primes is prime.*)

3.3 The Zariski spectrum and its topology

We have a set of points, $\text{Spec } R = \{\text{prime ideals of } R\}$. To do geometry we need a *topology*, and there is exactly one natural choice: the closed sets should be the places where collections of functions vanish. Recall from §3.2/§1 that f vanishes at $\mathfrak{p}$ iff $f \in \mathfrak{p}$.

Definition 3.4 (Vanishing locus and the Zariski topology)

For an ideal $I \subseteq R$ set

$$V(I) \stackrel{\text{def}}{=} \{\mathfrak{p} \in \text{Spec } R : I \subseteq \mathfrak{p}\} = \{\text{points where every } f \in I \text{ vanishes}\}.$$

The sets $V(I)$ are the **closed sets** of the **Zariski topology** on $\text{Spec } R$. Its basic open sets are the **distinguished opens** $D(f) \stackrel{\text{def}}{=} \text{Spec } R \setminus V(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$, the locus where f is nonzero.

That this is a topology is a short computation, and it is worth doing because each identity is a geometric statement.

Proposition 3.3 (The Zariski axioms)

$V(0) = \text{Spec } R$ and $V(R) = \emptyset$; for any family of ideals $\bigcap_{\alpha} V(I_{\alpha}) = V(\sum_{\alpha} I_{\alpha})$; and for two ideals $V(I) \cup V(J) = V(I \cap J) = V(IJ)$. Hence the $V(I)$ are closed under arbitrary intersection and finite union, so they are the closed sets of a topology. The $D(f)$ form a basis, with $D(f) \cap D(g) = D(fg)$.

Proof. The first two are immediate (0 lies in every prime; no prime contains 1). For intersections, $\mathfrak{p} \supseteq I_{\alpha}$ for all α iff $\mathfrak{p} \supseteq \sum_{\alpha} I_{\alpha}$. For unions: $IJ \subseteq I \cap J$ gives $V(I \cap J) \subseteq V(IJ)$, and $V(I) \cup V(J) \subseteq V(I \cap J)$ is clear; conversely if $\mathfrak{p} \supseteq IJ$ but $\mathfrak{p} \not\supseteq I$, pick $a \in I \setminus \mathfrak{p}$; then for every $b \in J$, $ab \in IJ \subseteq \mathfrak{p}$ forces $b \in \mathfrak{p}$ (primeness!), so $\mathfrak{p} \supseteq J$. Thus $V(IJ) \subseteq V(I) \cup V(J)$, and the three sets coincide. Finally $f \notin \mathfrak{p}$ and $g \notin \mathfrak{p}$ iff $fg \notin \mathfrak{p}$ (primeness again), giving $D(f) \cap D(g) = D(fg)$; that the $D(f)$ are a basis is because $V(I) = \bigcap_{f \in I} V(f)$, so its complement is $\bigcup_{f \in I} D(f)$.⁸

Notice where primeness was used: *exactly* in making $V(IJ) = V(I) \cup V(J)$ and $D(fg) = D(f) \cap D(g)$. The point-set geometry of $\text{Spec } R$ works *because* the points are primes—further evidence that this, not the maximal ideals, is the right notion of point.

⁸A–M Ch. 1, Ex. 15–17, where this is developed as the celebrated sequence of exercises that quietly introduces the language of schemes; A–K §§13, 16 (Support and Noetherian Spaces).

Intuition 3.2 (The Zariski topology is coarse on purpose)

Do not picture the metric topology. The Zariski topology is extremely *coarse*: closed sets are just common zero-loci of polynomials, so there are very few of them, and points are enormous. On $\text{Spec } \mathbb{Z}$ or $\text{Spec } k[x]$ the proper closed sets are the *finite* sets of closed points, so every two nonempty opens meet—the space is far from Hausdorff. The coarseness is a feature: it is the finest topology in which “polynomial” makes sense, and it is what lets a single *generic point* be dense.

Two structural features deserve names, because they recur constantly.

Generic points and irreducibility. The closure of a point is $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$: indeed $\mathfrak{q} \in \overline{\{\mathfrak{p}\}}$ iff every closed set containing $\mathfrak{p}$ contains $\mathfrak{q}$, i.e. iff $\mathfrak{p} \subseteq \mathfrak{q}$. So $\mathfrak{p}$ is dense in the irreducible closed set $V(\mathfrak{p})$, of which it is the **generic point**. When R is a domain, (0) is a generic point for *all* of $\text{Spec } R$: the domain’s spectrum has a single dense point, the visible sign of irreducibility. Dually, a point $\{\mathfrak{p}\}$ is *closed* exactly when $V(\mathfrak{p}) = \{\mathfrak{p}\}$, i.e. when $\mathfrak{p}$ is maximal—so the closed points are precisely the classical points, while a non-maximal prime is never closed. Distinct primes have distinct closures, so $\text{Spec } R$ is always a T_0 space, even though it is almost never T_1 or Hausdorff. In general the irreducible components of $\text{Spec } R$ are the $V(\mathfrak{p})$ for $\mathfrak{p}$ a minimal prime.⁹

Quasi-compactness. Unlike the metric topology, $\text{Spec } R$ is always **quasi-compact**: every open cover has a finite subcover.

Exercise 3.4 ((classic) Spec R is quasi-compact) — Show $\text{Spec } R$ is quasi-compact. (*Hint: it suffices to consider covers by basic opens $D(f_\alpha)$; $\bigcup_\alpha D(f_\alpha) = \text{Spec } R$ means the f_α lie in no common prime, i.e. generate the unit ideal, so $1 = \sum_{i=1}^n r_i f_{\alpha_i}$ is a finite relation—now finish.*)

Exercise 3.5 ((medium) when is a function invisible or everywhere nonzero?) — Show $D(f) = \emptyset$ iff f is nilpotent, and $D(f) = \text{Spec } R$ iff f is a unit. Interpret both geometrically. (These are the extreme cases of “the locus where $f \neq 0$.”)

3.4 Nilradical, Jacobson radical, and the radical of an ideal

The topology $V(I)$ cannot distinguish I from larger ideals with the same zero-locus, so we need to know *which* algebraic data the geometry remembers. The answer is governed by radicals, and the story begins with the functions that vanish *everywhere*.

Definition 3.5 (Nilradical; reduced rings)

The **nilradical** $\text{Nil}(R)$ is the set of **nilpotent** elements $\{f : f^n = 0 \text{ for some } n \geq 1\}$. A ring with $\text{Nil}(R) = 0$ is **reduced**.

Theorem 3.2 (The nilradical is the intersection of all primes)

⁹A–M Ch. 1, Ex. 19–20; $\text{Spec } R$ is irreducible iff the nilradical (§3.4) is prime.

$\text{Nil}(R) = \bigcap_{\mathfrak{p} \in \text{Spec } R} \mathfrak{p}$. Equivalently, the nilpotents are exactly the “functions” whose value is 0 at every point of $\text{Spec } R$.

Proof. ($\subseteq$) If $f^n = 0$ then $f^n \in \mathfrak{p}$ for every prime $\mathfrak{p}$, and primeness gives $f \in \mathfrak{p}$. ($\supseteq$) Suppose f is not nilpotent; we find a prime missing f . The set $\Sigma = \{1, f, f^2, \dots\}$ avoids 0, so by Zorn there is an ideal $\mathfrak{p}$ maximal among ideals disjoint from Σ . This $\mathfrak{p}$ is prime: if $a, b \notin \mathfrak{p}$ then $\mathfrak{p} + (a)$ and $\mathfrak{p} + (b)$ each meet Σ (by maximality), so some $f^m \in \mathfrak{p} + (a)$ and $f^n \in \mathfrak{p} + (b)$; multiplying, $f^{m+n} \in \mathfrak{p} + (ab)$, so $\mathfrak{p} + (ab)$ meets Σ and hence $ab \notin \mathfrak{p}$. Thus $f \notin \mathfrak{p} \supseteq$ the intersection.¹⁰

This is the precise form of the §1 slogan “a nonzero nilpotent vanishes at every point”—and it shows in passing that $\text{Nil}(R)$ really is an ideal, being an intersection of primes. The nilradical is the gap between R and the functions $\text{Spec } R$ can actually see; $R_{\text{red}} \stackrel{\text{def}}{=} R/\text{Nil}(R)$ is the reduced ring with the same spectrum (as a space).

Applying the theorem inside a quotient gives the general radical.

Definition 3.6 (Radical of an ideal)

The **radical** of an ideal I is $\sqrt{I} \stackrel{\text{def}}{=} \{f : f^n \in I \text{ for some } n \geq 1\}$. An ideal with $\sqrt{I} = I$ is a **radical ideal**.

Proposition 3.4 (Radical = intersection of the primes above I)

$\sqrt{I} = \bigcap_{\mathfrak{p} \supseteq I} \mathfrak{p}$, and consequently $V(I) = V(\sqrt{I})$. Thus radical ideals are exactly the ones recoverable from their vanishing locus, and $I \mapsto V(I)$ restricts to a bijection between radical ideals and Zariski-closed sets that “come from ideals.”

Proof. $\sqrt{I}/I = \text{Nil}(R/I) = \bigcap \{\text{primes of } R/I\}$, which under the correspondence theorem is $(\bigcap_{\mathfrak{p} \supseteq I} \mathfrak{p})/I$. Intersecting the same primes gives the same V , so $V(I) = V(\sqrt{I})$.¹¹

Example 3.4 (Computing radicals)

In $\mathbb{Z}$, $\sqrt{(12)} = (6)$ because $12 = 2^2 \cdot 3$ and the primes above (12) are (2), (3), with $(2) \cap (3) = (6)$. In $k[x, y]$, $\sqrt{(x^2 y^3)} = (xy)$, with vanishing locus the union of the two axes. In $k[x]/(x^2)$, $\text{Nil} = (x)$ and the reduced ring is k : the fat point’s underlying space is a single reduced point.

Finally, a sibling of the nilradical built from the *maximal* ideals instead of all primes.

Definition 3.7 (Jacobson radical)

The **Jacobson radical** $\text{Jac}(R) = \bigcap_{\mathfrak{m} \text{ maximal}} \mathfrak{m}$.

¹⁰A–M Ch. 1, Prop. 1.8; A–K §3. A slicker proof via localization (f not nilpotent $\Rightarrow R_f \neq 0 \Rightarrow R_f$ has a prime) waits for §6.

¹¹A–M Ch. 1, Prop. 1.14. This is the engine of the Nullstellensatz (§10), which for finitely generated algebras over $k = \bar{k}$ upgrades “ $\sqrt{I} = \bigcap \mathfrak{p}$ ” to “ $\sqrt{I} = I(V(I))$,” a perfect dictionary between radical ideals and varieties.

It has a clean elementary characterization, the one that makes it useful in Nakayama's lemma (§4): $x \in \text{Jac}(R) \iff 1 - xy$ is a unit for every $y \in R$.¹² In a local ring $\text{Jac}(R)$ is the unique maximal ideal $\mathfrak{m}$ itself; in $\mathbb{Z}$, $\text{Jac}(\mathbb{Z}) = \bigcap_p (p) = 0$, since no integer is divisible by every prime.

Exercise 3.6 ((easy) radicals and reducedness) — Compute $\sqrt{I}$ for $I = (9)$ and $I = (72)$ in $\mathbb{Z}$, and for $I = (x^3)$ in $k[x]$. Which of $\mathbb{Z}/72\mathbb{Z}$, $k[x]/(x^3)$ are reduced?

Exercise 3.7 ((thinker) nilpotents are invisible to the topology) — Show that $\text{Spec } R$ and $\text{Spec } R_{\text{red}}$ are homeomorphic, naturally. Conclude that the Zariski topology alone cannot detect nilpotents—this is precisely why *schemes* carry the whole ring R (a structure sheaf), not just the space (§15).

3.5 Spec as a contravariant functor: ring maps are space maps

We promised in §1 that algebra would translate to geometry automatically. Now we deliver: a ring map induces a continuous map of spectra, in the opposite direction, compatibly with composition. This makes Spec a *functor*, and turns the two most basic algebraic operations—quotient and localization—into the two most basic geometric ones, closed and open embeddings.

Theorem 3.3 (Spec is a contravariant functor)

A ring homomorphism $\varphi: R \rightarrow S$ induces a map

$$\varphi^* = \text{Spec}(\varphi): \text{Spec } S \longrightarrow \text{Spec } R, \quad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q}),$$

which is well defined (preimages of primes are prime, §1) and *continuous*, because $(\varphi^*)^{-1}(V(I)) = V(\varphi(I)S)$ for every ideal $I \subseteq R$. Moreover $(\psi \circ \varphi)^* = \varphi^* \circ \psi^*$ and $\text{id}^* = \text{id}$. Thus $\text{Spec}: \text{Ring} \rightarrow \text{Top}$ is a contravariant functor.

Proof. Continuity: $\mathfrak{q} \in (\varphi^*)^{-1}(V(I))$ iff $\varphi^{-1}(\mathfrak{q}) \supseteq I$ iff $\mathfrak{q} \supseteq \varphi(I)$ iff $\mathfrak{q} \supseteq \varphi(I)S$ (as $\mathfrak{q}$ is an ideal of S), i.e. $\mathfrak{q} \in V(\varphi(I)S)$. Functoriality is immediate from $(\psi \circ \varphi)^{-1} = \varphi^{-1} \circ \psi^{-1}$.¹³

The two cases to know are the ones that give geometry its closed and open pieces.

Equation 3.1 (Quotient = closed; localization = open)

$$\text{Spec}(R/I) \cong V(I) \subseteq \text{Spec } R \quad (\text{closed}); \quad \text{Spec}(R_f) \cong D(f) \subseteq \text{Spec } R \quad (\text{open}).$$

Imposing equations ($R \rightarrow R/I$) restricts to a closed subspace; inverting a function ($R \rightarrow R_f$) restricts to its open nonvanishing locus.

The closed case is the correspondence theorem of §3.1 in disguise: primes of R/I are primes of R containing I , i.e. the points of $V(I)$, and the subspace topology matches. The open case is

¹²A–M Ch. 1, Prop. 1.9. Always $\text{Nil}(R) \subseteq \text{Jac}(R)$; they are equal for rings finitely generated over a field (**Jacobson rings**), where the maximal ideals are already dense enough to detect nilpotents—a foretaste of the Nullstellensatz.

¹³A–M Ch. 1, Ex. 21; A–K §13. The operation $I \mapsto \varphi(I)S$ is the **extension** of I , written I^e ; pulling back is **contraction**, $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q}) = \mathfrak{q}^c$.

the content of localization and is proved in §6; we state it here so the symmetry is visible now. One more general fact ties a map's reach to its kernel: the closure of the image of φ^* is exactly $V(\ker \varphi)$. Indeed that closure is the smallest $V(I)$ containing the image, namely $V(\bigcap_{\mathfrak{q}} \varphi^{-1}(\mathfrak{q}))$, and $\bigcap_{\mathfrak{q}} \varphi^{-1}(\mathfrak{q}) = \varphi^{-1}(\text{Nil } S) = \sqrt{\ker \varphi}$ has the same vanishing locus as $\ker \varphi$. So φ^* has dense image precisely when $\ker \varphi \subseteq \text{Nil}(R)$ —in particular whenever φ is injective, though injectivity is not necessary.¹⁴

Example 3.5 (Ring maps as geometry) • **Structure map $k \rightarrow k[x]$.** Induces $\mathbb{A}^1 = \text{Spec } k[x] \rightarrow \text{Spec } k = \text{pt}$: the affine line sits over the point, as *every* k -algebra sits over $\text{Spec } k$ (and every ring over $\text{Spec } \mathbb{Z}$, §2).

- **Inclusion $k[x] \hookrightarrow k[x, y]$.** Induces the projection $\mathbb{A}^2 \rightarrow \mathbb{A}^1$, $(a, b) \mapsto a$; the fiber over a point of the line is a vertical line, exactly the primes of $k[x, y]$ contracting to $(x - a)$.
- **$\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$.** Induces the branched double cover $\text{Spec } \mathbb{Z}[i] \rightarrow \text{Spec } \mathbb{Z}$ of §1; the fiber over (p) encodes how p splits, ramifies, or stays inert. This is the geometry of $\mathcal{O}_K \rightarrow \mathcal{O}_K$ over $\text{Spec } \mathbb{Z}$ that runs algebraic number theory.

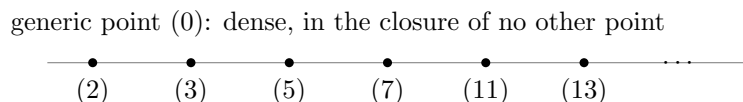
Note 3.2 (The anti-equivalence at the heart of algebraic geometry). Everything in this subsection is the shadow of one clean statement: Spec is (the object part of) an *anti-equivalence* between commutative rings and affine schemes, $\text{Ring}^{\text{op}} \simeq \text{AffSch}$. Reversing arrows is not a quirk—it is the same contravariance as “functions pull back” (§1). Gluing these affine pieces along distinguished opens produces general **schemes** (§15), and the terminal object $\text{Spec } \mathbb{Z}$ (§2) is the base over which all of arithmetic geometry lives.

Exercise 3.8 ((classic) functoriality and the quotient) — Prove $(\psi \circ \varphi)^* = \varphi^* \circ \psi^*$ in detail. Then show directly that for the quotient $\pi: R \rightarrow R/I$, the map π^* is a homeomorphism of $\text{Spec}(R/I)$ onto the closed set $V(I)$.

3.6 First pictures: the spectra of $\mathbb{Z}$, $k[x]$, $k[x, y]$, and a fat point

Time to draw. The pictures below are the ones every algebraic geometer carries in their head; sketching them yourself is the fastest way to make the dictionary automatic.

$\text{Spec } \mathbb{Z}$: the arithmetic line. One closed point (p) for each prime number, with residue field $\mathbb{F}_p$, plus a single generic point (0) (residue field $\mathbb{Q}$) whose closure is everything. It looks like a discrete line of dots—except for the omnipresent generic point smeared over all of them.

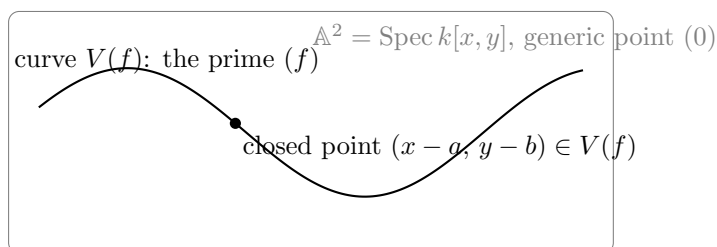


$\text{Spec } k[x]$: the affine line $\mathbb{A}^1$. For $k = \bar{k}$ the closed points $(x - a)$ are the field k , with residue field k and value $f \mapsto f(a)$, plus the generic point (0) . The structural twin of $\text{Spec } \mathbb{Z}$ is exact—one generic point over a line of closed points—which is the $\mathbb{Z} \leftrightarrow k[x]$ analogy of §1 made visible.

¹⁴A–M Ch. 1, Ex. 21(v). The quotient $R \rightarrow R_{\text{red}}$ is the extreme case: not injective, yet a homeomorphism on spectra.

Remark 3.1 (Closed points over a non-closed field are Galois orbits). If $k \neq \bar{k}$ the closed points of $\text{Spec } k[x]$ are the *irreducible* monic polynomials, not the elements of k . Over $\mathbb{R}$, the point $(x^2 + 1)$ has residue field $\mathbb{R}[x]/(x^2 + 1) = \mathbb{C}$ and corresponds to the conjugate pair $\{i, -i\}$: a single Zariski point that is a *Galois orbit* $\{a, \bar{a}\}$ of geometric points. Thus $\text{Spec } \mathbb{R}[x]$ already encodes the action of $\text{Gal}(\mathbb{C}/\mathbb{R})$ —the first whisper of how Galois theory and geometry merge in arithmetic, and of the Galois side of the Langlands correspondence.

$\text{Spec } k[x, y]$: the affine plane $\mathbb{A}^2$. Now there are three tiers of points, nested by the chain of primes $(0) \subsetneq (f) \subsetneq (x - a, y - b)$: the generic point (0) (the whole plane), the curve-points (f) for f irreducible (each the generic point of its curve $V(f)$), and (taking $k = \bar{k}$) the closed points $(x - a, y - b)$, the honest points of the plane. A maximal ideal $(x - a, y - b)$ contains (f) exactly when the point (a, b) lies on the curve $f = 0$ —so inclusion of primes is incidence of geometry.



This picture is also your first sight of **dimension**: the longest chain of primes has length two, and $\dim k[x, y] = 2$ (§12). Counting how many times you can nest irreducible closed sets is counting dimensions.¹⁵

A fat point: $\text{Spec } k[x]/(x^2)$. The ring $k[x]/(x^2)$ has the single prime (x) , so its spectrum is *one point*—the same space as $\text{Spec } k$. But the rings differ: $k[x]/(x^2)$ carries the nonzero nilpotent x . Geometrically this is a point with an infinitesimal arrow attached, the **dual numbers** $k[\varepsilon]/(\varepsilon^2)$, whose maps to it are exactly tangent vectors.¹⁶ This is the payoff of the §1 nilpotent discussion: two schemes can share a topological space yet differ as rings, which is why the space alone is not enough and the ring must be remembered.

Example 3.6 (A reducible spectrum: two lines crossing)

$\text{Spec } k[x, y]/(xy)$ is the union of the two coordinate axes $V(x) \cup V(y)$. Its minimal primes are (x) and (y) —the two irreducible components—and they meet at the closed point (x, y) , the origin. The ring is reduced ($\sqrt{(xy)} = (xy)$) but not a domain, the algebraic signature of a space that is a union of irreducible pieces.

Exercise 3.9 ((medium) draw a cover) — Sketch $\text{Spec } \mathbb{Z}[i] \rightarrow \text{Spec } \mathbb{Z}$ over the primes 2, 3, 5, 7, 11, 13, marking which are split (two points in the fiber), inert (one point), or ramified.

¹⁵The arithmetic analogue is $\text{Spec } \mathbb{Z}[x]$, a two-dimensional “arithmetic surface” fibered over $\text{Spec } \mathbb{Z}$ (A–M Ch. 1, Ex. 16). Its hand-drawn portrait—Mumford’s famous picture—mixes vertical primes (p) with horizontal curves and is the iconic image of arithmetic geometry; we return to it in §15.

¹⁶A map of k -algebras $k[\varepsilon]/(\varepsilon^2) \rightarrow A$ is a point of $\text{Spec } A$ together with a tangent vector there; this is how schemes do calculus. See §15 and, e.g., Eisenbud–Harris, *The Geometry of Schemes*.

Relate your picture to $p \equiv 1, 3 \pmod{4}$.

Exercise 3.10 ((classic) disconnected spectra and idempotents) — Show $\operatorname{Spec}(R \times S) \cong \operatorname{Spec} R \sqcup \operatorname{Spec} S$ (a disjoint union of spaces), so a product ring has a disconnected spectrum. Conversely, show $\operatorname{Spec} R$ is disconnected iff R has a nontrivial idempotent $e = e^2 \notin \{0, 1\}$, which then splits $R \cong Re \times R(1 - e)$. (*This is the algebra of a space falling into two clopen pieces.*)

Exercise 3.11 ((thinker) the spectrum remembers more than the space) — Exhibit two rings with homeomorphic spectra that are not isomorphic, and two non-isomorphic rings whose spectra differ only by a nilpotent “thickening.” Explain what extra data a scheme records beyond $\operatorname{Spec} R$ as a topological space.

3.7 Connections to further mathematics and physics

You now hold the central object of the subject, $\operatorname{Spec} R$, with its topology, its radicals, and its functoriality. This is the launchpad for a great deal of mathematics and physics.

Within these notes and mathematics. Modules become **quasi-coherent sheaves** on $\operatorname{Spec} R$ (§4, §15); localization carves out the open sets that let us *glue* affine spectra into **schemes** (§6, §15); the chains of primes you saw in $k[x, y]$ become **Krull dimension** (§12); and the maximal-ideal/point correspondence becomes the **Nullstellensatz** (§10). In number theory $\operatorname{Spec} \mathcal{O}_K$ is the basic object of arithmetic geometry, and the arithmetic surface $\operatorname{Spec} \mathbb{Z}[x]$ is where Diophantine geometry lives—so close to a complex surface that one studies it with the same intersection theory (Arakelov geometry).

In physics. The Zariski geometry of $\operatorname{Spec} R$ is, concretely, the geometry of vacua.

Idea 3.1 (Branches, singularities, and nilpotents are physical)

For a supersymmetric field theory whose chiral ring is R , the moduli space of vacua is $\operatorname{Spec} R$ (the §1 connection), and the structures of this section are its physics:

- *irreducible components* ($= V(\text{minimal primes})$, §3.3) are the distinct **branches** of vacua—the Coulomb, Higgs, and mixed branches of a gauge theory;
- the *singular points* where branches meet—like the origin of $\operatorname{Spec} k[x, y]/(xy)$ —are loci of *enhanced gauge symmetry* or extra massless states, the conifold being the famous example;
- *nilpotents* in R record infinitesimal directions and obstructions the naive vacuum manifold forgets, exactly the data a scheme keeps and a topological space loses (§3.4).

Reading branches and singularities off primes and radicals is a daily tool in the study of gauge theories.^a

^aFor branches of moduli space and prime decomposition, see Argyres’s lectures; for D-branes at singularities, where the *noncommutative* coordinate ring’s Spec resolves the singularity into a **quiver** gauge theory, see Douglas–Moore.

The same picture drives the AdS/CFT correspondence—the geometry probed by branes is the Spec (or Proj) of a ring built from gauge-invariant operators—and string compactification, where Calabi–Yau singularities and their resolutions are studied through the commutative algebra of their

coordinate rings. The arithmetic line and the moduli space of a gauge theory are two faces of the single construction Spec .

3.8 Exercises

The richest classical problem set in the subject is Chapter 1 of Atiyah–Macdonald, mirrored in Altman–Kleiman §§1–3. The following are among the most instructive; each drills a reflex you will use constantly.

Exercise 3.12 ((classic) units and nilpotents in a polynomial ring (A–M Ch. 1, Ex. 2)) — Let $f = a_0 + a_1x + \cdots + a_nx^n \in R[x]$. Prove: (i) f is a unit $\iff a_0$ is a unit and $a_1, \dots, a_n$ are nilpotent; (ii) f is nilpotent $\iff$ every a_i is nilpotent; (iii) f is a zero-divisor $\iff cf = 0$ for a single $c \neq 0$ in R . (Part (i) is a genuine surprise: units in $R[x]$ exceed the constants exactly when R has nilpotents.)

Exercise 3.13 ((medium) rings of dimension zero (A–M Ch. 1, Ex. 7; A–K 2.38)) — Suppose every $x \in R$ satisfies $x^n = x$ for some $n > 1$ (depending on x). Show every prime of R is maximal—so every point of $\text{Spec } R$ is closed and $\dim R = 0$. (Boolean rings are the case $n = 2$.)

Exercise 3.14 ((classic) Boolean rings and Stone duality (A–M Ch. 1, Ex. 11; A–K 2.37)) — A ring is **Boolean** if $x^2 = x$ for all x . Show (i) $2x = 0$ for all x ; (ii) every prime is maximal with residue field $\mathbb{F}_2$; (iii) every finitely generated ideal is principal. (Then $\text{Spec } R$ is compact and totally disconnected, and $R \mapsto \text{Spec } R$ is **Stone duality** between Boolean algebras and such spaces—the bridge from logic to topology, and a baby case of the Spec anti-equivalence.)

Exercise 3.15 ((medium) zero-divisors are a union of primes (A–M Ch. 1, Ex. 14; A–K 2.26)) — Let Σ be the set of ideals all of whose elements are zero-divisors. Show Σ has maximal elements (Zorn) and that each is prime. Conclude the set of zero-divisors of R is a union of primes. (These are the **associated primes** of §11—your first sighting.)

Exercise 3.16 ((classic) prime avoidance, the core case (A–K 2.23)) — For ideals $\mathfrak{a}, \mathfrak{b}$ and a prime $\mathfrak{p}$, prove the equivalence of: (1) $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$; (2) $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$; (3) $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$. (The reflex “a prime swallowing a product/intersection swallows a factor”—used everywhere, and the seed of the general prime-avoidance lemma.)

Exercise 3.17 ((medium) when is the spectrum irreducible? (A–M Ch. 1, Ex. 19–20)) — Show $\text{Spec } R$ is irreducible (every two nonempty opens meet) iff $\text{Nil}(R)$ is prime, and that the irreducible components of $\text{Spec } R$ are the $V(\mathfrak{p})$ for $\mathfrak{p}$ a minimal prime. (“How many pieces the space has” = “how many minimal primes R has.”)

Exercise 3.18 ((thinker) a domain bounds its roots; a non-domain need not (A–K 2.35)) — Show a degree- d polynomial over a *domain* has at most d roots. Then check that $X^2 + X$ has *four* roots in $\mathbb{Z}/6\mathbb{Z}$. Explain, using $\mathbb{Z}/6 \cong \mathbb{F}_2 \times \mathbb{F}_3$ (CRT) and the two-point space $\text{Spec}(\mathbb{Z}/6)$, why the bound fails off a domain. (*Zero-divisors break factor-counting; the disconnected spectrum is the reason.*)

Read alongside: A–M Ch. 1 is the spine of this section: the body for ideals, primes, maximals, and the two radicals (Props. 1.8, 1.9, 1.14), and the remarkable Exercises 15–21 for the Zariski spectrum, its topology, quasi-compactness, irreducible components, and functoriality. A–K §§1–3 redo this with the modern emphasis, and §13 (Support) and the Appendix to §16 (Noetherian Spaces) carry the topology further. Aluffi Ch. III §§3–4 is the gentlest source for ideals and quotient rings. For the geometric reading, Eisenbud–Harris, *The Geometry of Schemes*, Ch. I, is the ideal companion.

4 Modules: linear algebra over a ring

A ring is best understood through the things it acts on, and those are its **modules**. Modules are “vector spaces over a ring,” and the slogan of the modern subject is that commutative algebra is the study of the category $R\text{-Mod}$. Almost everything we already know is a module: ideals, quotients, abelian groups (= $\mathbb{Z}$ -modules), vector spaces (= modules over a field). The one new phenomenon—and the source of the whole subject’s richness—is that *not every module is free*; measuring that failure is what occupies us for the rest of the notes. We introduce annihilators and support to begin seeing modules as objects spread out over $\text{Spec } R$.

- 4.1 Modules and the category $R\text{-Mod}$: why everything is a module
- 4.2 Submodules, quotients, Hom , and the operations on submodules
- 4.3 Generators and relations; finitely generated and finitely presented modules
- 4.4 Free modules, and the productive failure of “every module is free”
- 4.5 Annihilators and support: a first glimpse of modules as sheaves on $\text{Spec } R$
- 4.6 Connections to further mathematics and physics
- 4.7 Exercises

Read alongside: A–M Ch. 2; A–K §4 (Modules); Aluffi Ch. III §5 and Ch. VI.

5 Exact sequences and the homological reflex

Exact sequences are how modern algebra bookkeeps relationships between modules, and learning to read them fluently is half the battle. We cover short and long exact sequences, the **snake lemma** (the workhorse of every diagram chase), and—crucially—the idea that a functor’s *failure* to preserve exactness is information, not a nuisance. That single reframing is what turns Hom and $\otimes$ into the derived functors Ext and Tor later, and it is the reflex that distinguishes the modern treatment from the classical one.

- 5.1 Kernels, cokernels, images; short and long exact sequences
- 5.2 The snake lemma and the art of the diagram chase
- 5.3 Extensions and splitting: what an exact sequence remembers
- 5.4 Additive, left-exact, and right-exact functors as a measuring stick
- 5.5 Connections to further mathematics and physics
- 5.6 Exercises

Read alongside: A–M Ch. 2 (exact sequences); A–K §5; Aluffi Ch. VI §7 and Ch. VIII–IX.

6 Localization: zooming in on the spectrum

Localization is the most important *technique* in the subject, and intuition is the whole battle, so it gets room to breathe. Geometrically, inverting a set of elements means *zooming in* on part of $\operatorname{Spec} R$; in particular, localizing at a prime $\mathfrak{p}$ produces the **local ring** $R_{\mathfrak{p}}$, which records exactly the behavior of R near the point $\mathfrak{p}$. We give both the universal property and the explicit fractions construction, say why they describe the same object, and then state the **local–global principle**: a great many properties hold for R if and only if they hold at every prime. That principle is the engine that lets us prove global facts one point at a time.

- 6.1 The problem: studying a ring near a point, or inverting a set of elements
- 6.2 The construction $S^{-1}R$ and $S^{-1}M$, and its universal property
- 6.3 The two cases that matter: a basic open R_f and a local ring $R_{\mathfrak{p}}$
- 6.4 Local rings, residue fields, and why “local” is the right word
- 6.5 Localization is exact; the order-preserving correspondence of primes
- 6.6 The local–global principle: prove it at every prime
- 6.7 Connections to further mathematics and physics
- 6.8 Exercises

Read alongside: A–M Ch. 3 (this is the heart of the book—read it twice); A–K §§11–13 (Localization of Rings, of Modules, and Support).

7 Tensor products, base change, and flatness

The tensor product is where the categorical language earns its keep. We give the universal property (the clean definition) *and* the generators-and-relations construction (the one you compute with), and we say precisely why they are the same object. The key structural fact is **adjoint associativity**: $-\otimes_R N$ is left adjoint to $\operatorname{Hom}_R(N, -)$, from which right-exactness falls out for free. **Flatness**—the modules for which tensoring preserves exactness—is the algebraic shadow of “continuously varying

family,” and it is one of the deepest geometric ideas in the subject. We close with Lazard’s theorem, the modern characterization of flat as a filtered colimit of finite free modules.

- 7.1 Bilinear maps, the universal property, and the explicit construction (why they agree)**
- 7.2 Tensor as a functor; right-exactness from adjoint associativity**
- 7.3 Extension of scalars and base change; localization is base change**
- 7.4 Flatness: the modules that preserve exactness, i.e. “continuous families”**
- 7.5 Lazard’s theorem: flat = filtered colimit of finite free modules**
- 7.6 Connections to further mathematics and physics**
- 7.7 Exercises**

Read alongside: A–M Ch. 2–3; A–K §§6–9 (Direct Limits, Filtered Direct Limits, Tensor Products, Flatness)—the most thoroughly modern treatment of the three; Aluffi Ch. VIII §2.

8 Derived functors: Tor and Ext

Once we accept that $\otimes$ and Hom fail to be exact, the natural question is: by how much, and what does the failure measure? The answer is **homological algebra**. We build projective and injective resolutions, define **Tor** (the derived functor of $\otimes$) and **Ext** (the derived functor of Hom), and read their meaning: Tor detects flatness and controls how subvarieties intersect, while Ext^1 classifies extensions and is the home of obstructions. This is the toolkit behind depth, regular sequences, and the homological characterization of smoothness that reappears in the dimension theory below.

- 8.1 Projective and injective modules, and resolutions**
- 8.2 Tor: the homology of $\otimes$, and what its vanishing means**
- 8.3 Ext: extensions, obstructions, and Ext^1 as a classifying group**
- 8.4 Homological algebra as the modern measure of “how far from exact”**
- 8.5 Connections to further mathematics and physics**
- 8.6 Exercises**

Read alongside: A–K (the flatness and Tor threads in §9 and beyond); Aluffi Ch. VIII §§2–6 and Ch. IX (Homological algebra). For depth and the Auslander–Buchsbaum circle, see Eisenbud later.

9 Integral extensions and the geometry of finite maps

Integral extensions are the algebraic side of *finite maps* of spaces—think of a branched cover of curves. We develop integral elements, integral closure, and **normal** rings, then the three transfer theorems—**lying over**, **going up**, and **going down**—which say how primes (points) move under a

finite map. **Noether normalization**, the statement that every finitely generated algebra over a field is finite over a polynomial subring, is both a structural cornerstone and the lever we use to prove the Nullstellensatz in the next section.

- 9.1 Integral elements, integral extensions, integral closure, and normal rings
- 9.2 Lying over, going up, going down: how points move under a finite map
- 9.3 Noether normalization: every affine algebra is finite over a polynomial ring
- 9.4 Geometric meaning: finite surjections, branching, and normalization
- 9.5 Connections to further mathematics and physics
- 9.6 Exercises

Read alongside: A–M Ch. 5; A–K §10 (Cayley–Hamilton—the slick route to integrality), §14 (Cohen–Seidenberg, i.e. going up/down), §15 (Noether Normalization).

10 The Nullstellensatz: the dictionary made precise

Here the algebra–geometry dictionary becomes a theorem. **Hilbert’s Nullstellensatz** identifies the maximal ideals of $k[x_1, \dots, x_n]$ (for k algebraically closed) with the points of affine space, and upgrades the loose correspondence “ideals $\leftrightarrow$ vanishing sets” into the precise statement $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. The upshot is an honest equivalence between affine varieties and reduced finitely generated k -algebras—the prototype of the ring/space duality and the doorway to scheme theory.

- 10.1 Hilbert’s Nullstellensatz, weak and strong
- 10.2 The radical–variety correspondence $\sqrt{\mathfrak{a}} = I(V(\mathfrak{a}))$
- 10.3 Coordinate rings and the equivalence varieties $\simeq$ reduced f.g. algebras
- 10.4 Why this is the gateway from varieties to schemes
- 10.5 Connections to further mathematics and physics
- 10.6 Exercises

Read alongside: A–M Ch. 5 and Ch. 7; A–K §15 (Noether Normalization gives the cleanest proof); field-theory background in Aluffi Ch. VII.

11 Chain conditions: Noetherian rings and primary decomposition

Almost every ring that arises in geometry or number theory satisfies the **ascending chain condition**—it is **Noetherian**—and **Hilbert’s basis theorem** guarantees this is preserved by the constructions we care about. The structural payoff is **primary decomposition**, the ring-theoretic analogue of factoring an integer, which we present *the modern way*: **associated primes** of a module from the very start (not, as in older books, extracted afterward from a decomposition). The associated

primes, together with the support, encode the irreducible components and the embedded points of a scheme—geometry read straight off the algebra.

- 11.1 Noetherian and Artinian modules; the ascending chain condition**
- 11.2 Hilbert’s basis theorem: why everything in sight is Noetherian**
- 11.3 Associated primes, defined from the start (the modern definition)**
- 11.4 Primary decomposition of modules and the uniqueness theorems**
- 11.5 Ass, Supp, and the geometry of components and embedded points**
- 11.6 Connections to further mathematics and physics**
- 11.7 Exercises**

Read alongside: A–M Ch. 4 (primary decomposition), Ch. 6–7 (chain conditions, Noetherian rings); A–K §§16–18 (Chain Conditions, Associated Primes, Primary Decomposition)—note A–K puts associated primes *first*, as is now standard.

12 Krull dimension: counting the directions a space can go

How many independent directions does a space have? The algebraic answer is **Krull dimension**: the longest chain of prime ideals. The deep and satisfying theorem is that three *a priori* different notions of dimension—chains of primes, the growth rate of the **Hilbert function**, and the minimal number of parameters cutting out a point—all agree for Noetherian local rings. **Krull’s principal ideal theorem** pins down codimension one, and **regular local rings** give the first clean algebraic criterion for *smoothness* versus singularity.

- 12.1 Krull dimension via chains of primes; height and codimension**
- 12.2 Hilbert functions, Hilbert polynomials, and dimension by growth**
- 12.3 Systems of parameters and the dimension theorem (the three notions agree)**
- 12.4 Krull’s principal ideal theorem and codimension one**
- 12.5 Regular local rings: a first algebraic criterion for smoothness**
- 12.6 Connections to further mathematics and physics**
- 12.7 Exercises**

Read alongside: A–M Ch. 11 (Dimension Theory); A–K §§19–21 (Length, Hilbert Functions, Dimension).

13 Completion: the local-analytic picture

Localization zooms in to a point; **completion** goes one step further and records the *infinitesimal, analytic* neighborhood of that point. Built as an inverse limit $\hat{R} = \varprojlim R/I^n$, completion produces objects like the p -adic integers $\mathbb{Z}_p$ and the formal power series $k[[x]]$, where convergence is free and

Hensel's lemma lets us lift approximate solutions to exact ones. Completion is a flat base change, so it imports the homological machinery intact, and the associated graded ring links it back to the Hilbert-function dimension theory just developed.

13.1 Filtrations, the I -adic topology, and inverse limits

13.2 Examples: $\mathbb{Z}_p$, $k[[x]]$, and formal neighborhoods

13.3 Hensel's lemma and lifting solutions; why complete local rings are “analytic”

13.4 Completion as a flat base change; the associated graded ring

13.5 Connections to further mathematics and physics

13.6 Exercises

Read alongside: A–M Ch. 10 (Completions; graded rings and the Artin–Rees lemma); A–K §22.

14 Dimension one: DVRs, Dedekind domains, fractional ideals

The one-dimensional case is where commutative algebra pays its great debt to number theory. A **discrete valuation ring** is the algebra of a single smooth point or a single prime, and a **Dedekind domain** is a ring—like the ring of integers of a number field—in which *ideals* factor uniquely even when *elements* do not. **Fractional ideals** and the **ideal class group** measure precisely how badly unique factorization fails, turning the classical headache of $\mathbb{Z}[\sqrt{-5}]$ into a clean invariant. This is the concrete arithmetic payoff of the whole machine.

14.1 Discrete valuation rings: the algebra of one smooth prime

14.2 Dedekind domains: unique factorization of ideals, not elements

14.3 Fractional ideals and the ideal class group

14.4 Rings of integers, $\mathbb{Z}[\sqrt{-5}]$, and the arithmetic payoff

14.5 Connections to further mathematics and physics

14.6 Exercises

Read alongside: A–M Ch. 9 (DVRs and Dedekind Domains); A–K §§23–26 (Discrete Valuation Rings, Dedekind Domains, Fractional Ideals, Arbitrary Valuation Rings).

15 Where this goes: schemes, sheaves, and the Langlands programs

A short outlook on what the path opens onto, with no claim to completeness. Gluing the affine pieces $\operatorname{Spec} R$ along their localizations gives **schemes**; promoting modules to $\operatorname{Spec} R$ globally gives **quasi-coherent sheaves** and their cohomology—this is the launchpad for algebraic geometry proper. On the arithmetic side, viewing $\operatorname{Spec} \mathbb{Z}$ as a “curve” and a ring of integers as a branched cover of it is the geometric heart of class field theory and, beyond it, the global, local, and geometric

Langlands programs, where every tool we built (localization, completion, Dedekind domains, Spec) is load-bearing.

15.1 From Spec to schemes: gluing affines and the structure sheaf

15.2 Quasi-coherent sheaves: modules globalized, and their cohomology

15.3 The arithmetic horizon: number fields and $\text{Spec } \mathbb{Z}$ as a curve

15.4 A word on the Langlands programs (global, local, geometric) and where CA sits

15.5 Connections to further mathematics and physics

15.6 Exercises

Read alongside: Outlook only—see the “what to read next” list: Hartshorne and Vakil for schemes, Neukirch for the arithmetic, and the survey literature for Langlands.

16 Capstone: problems that cross sections

Each section above ends with its own graded exercises drawn from Atiyah–Macdonald and Altman–Kleiman; do those as you go. This final set is different in kind—a handful of *synthesis* problems that deliberately span several sections at once, the sort that test whether the pieces have fused into a single picture. Attempt them after a full read; the “thinkers” push toward the references.

Exercise 16.1 ((easy) reading the dictionary) — For $R = k[x, y]/(xy)$, describe $\text{Spec } R$ as a topological space, list its minimal primes, and explain—using only the radical—why R is reduced but not a domain. Then say, in one sentence each, what “reduced” and “not a domain” mean *geometrically*.

Exercise 16.2 ((classic) localization is base change) — Let $S \subseteq R$ be multiplicative and M an R -module. Construct a natural isomorphism $S^{-1}M \cong S^{-1}R \otimes_R M$ and use it, together with right-exactness of $\otimes$, to re-prove that localization is exact. Where, exactly, does flatness of $S^{-1}R$ enter?

Exercise 16.3 ((medium) the local–global reflex) — Show that an R -module homomorphism $f: M \rightarrow N$ is injective (resp. surjective, resp. an isomorphism) if and only if $f_{\mathfrak{p}}: M_{\mathfrak{p}} \rightarrow N_{\mathfrak{p}}$ is for every prime $\mathfrak{p}$. State carefully which direction needs M or N finitely generated, and give a one-line geometric gloss of what the statement says about sections over $\text{Spec } R$.

Exercise 16.4 ((medium) dimension three ways, on one example) — For the local ring $R = k[x, y]_{(x, y)}$, compute the Krull dimension by chains of primes, by the degree of the Hilbert–Samuel polynomial, and by exhibiting a system of parameters. Verify all three give 2, and identify the maximal ideal’s minimal number of generators to check that R is regular.

Exercise 16.5 ((thinker) where unique factorization goes) — In $R = \mathbb{Z}[\sqrt{-5}]$, exhibit the failure of unique factorization on 6, then *repair* it: factor the ideal (6) into prime ideals and identify the relevant ideal classes. Explain how the class group converts a defect of *elements* into an invariant of the *ring*, and point to where Dedekind domains made this work. *This is the gateway to algebraic number theory; aim to internalize the mechanism, not just the computation.*

References and a reading plan

These notes are written against three primary sources, listed here in the order I would reach for them. Together they *are* a reading plan: A–M for the lean classical spine, A–K for the same material redone in fully modern categorical language, and Aluffi for the gentle, example-rich on-ramp to the categorical and module-theoretic ideas.

Primary sources (the three this path is built on).

- **M. F. Atiyah & I. G. Macdonald**, *Introduction to Commutative Algebra*. The standard concise reference; the entire subject in 130 pages, with the famous exercises. Terse but never wrong—read it with pen in hand and fill the gaps yourself. Best for the lean logical spine (§§3–14 of this path).
- **A. Altman & S. Kleiman**, *A Term of Commutative Algebra* (freely available). A–M rewritten for the modern era: associated primes from the start, direct limits and adjoints front and center, every result tied to a universal property, and complete solutions to every exercise. The best single match for the “modern, categorical” framing of these notes (especially §§2, 6–8).
- **P. Aluffi**, *Algebra: Chapter 0*. Categorical and unusually readable; the right place to learn universal properties, the module/linear-algebra perspective, and Tor/Ext without prior homological algebra. Best for §§2, 4–5, 8.

What to read next / books to own.

- **D. Eisenbud**, *Commutative Algebra with a View Toward Algebraic Geometry*. The big modern book: everything here, plus homological depth, regular sequences, and the geometry made explicit. The one to grow into.
- **H. Matsumura**, *Commutative Ring Theory*. The standard advanced reference for dimension theory, completion, and excellent rings.
- **M. Reid**, *Undergraduate Commutative Algebra*. A short, friendly, geometry-forward companion to read in parallel with A–M.
- **R. Hartshorne**, *Algebraic Geometry* (Ch. II) and **R. Vakil**, *The Rising Sea: Foundations of Algebraic Geometry* (math.stanford.edu/~vakil/216blog/). Where §15 actually goes: schemes and quasi-coherent sheaves.
- **J. Neukirch**, *Algebraic Number Theory*. The arithmetic continuation of §14—Dedekind domains, ramification, and $\text{Spec } \mathbb{Z}$ as a curve.
- **The Stacks Project** (stacks.math.columbia.edu). A complete, hyperlinked, ground-up development of commutative algebra and algebraic geometry; the modern reference of record.